Hydrological Flood Risk Mapping and Infrastructure Vulnerability Assessment Using Remote Sensing and Machine Learning in Hong Kong

Report

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Submission Format

Submit your work as either a GitHub repository (recommended) or a compressed ZIP archive. Your project structure should follow this format:

```
/scripts/          → All Python code
/maps/             → Final exported maps (PDFs)
/data/raw/         → Download links or metadata (do not include large raw
files)
/data/processed/   → Cleaned and clipped datasets
/qgis_project/     → Final QGIS project file (.qgz)
/report/           → Final report (PDF or Markdown)
README. md          → Tools used, instructions, and how to run your code
```

Solutions

1. Data Discovery and Evaluation –

You must independently acquire all datasets needed for the project. These may include (but are not limited to):

- **Hong Kong administrative boundary**
- **A Digital Elevation Model (DEM) with 30 m or better resolution**
- **Sentinel-2 imagery from 2022 or 2023 with minimal cloud cover**
- **Infrastructure layers: roads, hospitals, utilities, buildings**
- **Historical floodplain vector data**
- **Optional: legacy land use raster (for accuracy comparison)**

In your report, provide justification for each dataset selected. Clearly state the source, resolution, suitability, and any limitations or caveats.

Table 1 Datasets Details

S.No	Dataset	Raster/ Vector	Data and Links	Resolution
1.	Hong Kong SAR Boundary Shapefile	Vector	https://gadm.org/download_country.html	NA
2.	Sentinel 2 Data	Raster	https://browser.dataspace.copernicus.eu	10 m
3.	DEM Data - ASTER	Raster	https://search.earthdata.nasa.gov/search/granules?p=C1711961296-LPCLOUD&pg[0][v]=f&pg[0][gsk]=-start_date&q=aster&sb[0]=113.27555%2C21.71002%2C114.6546%2C22.87739&tl=1168862400!5!!&lat=22.26998808665362&long=114.25769082481013&zoom=8.601069930567078	30 m
4.	Infrastructure Data	Vector	Railway Lines, Education Facilities, Health Facilities - https://data.humdata.org/dataset/?q=hong+kong&ext_search_source=main-nav&page=2	NA

5.	Historical floodplain data	Raster	2006, 2007 and 2014 Hong Kong Flood Dataset - https://global-flood-database.cloudtostreet.ai/#interactive-map	250 m

Study Area

Hong Kong SAR, China (22.402 N, 114.119 E)

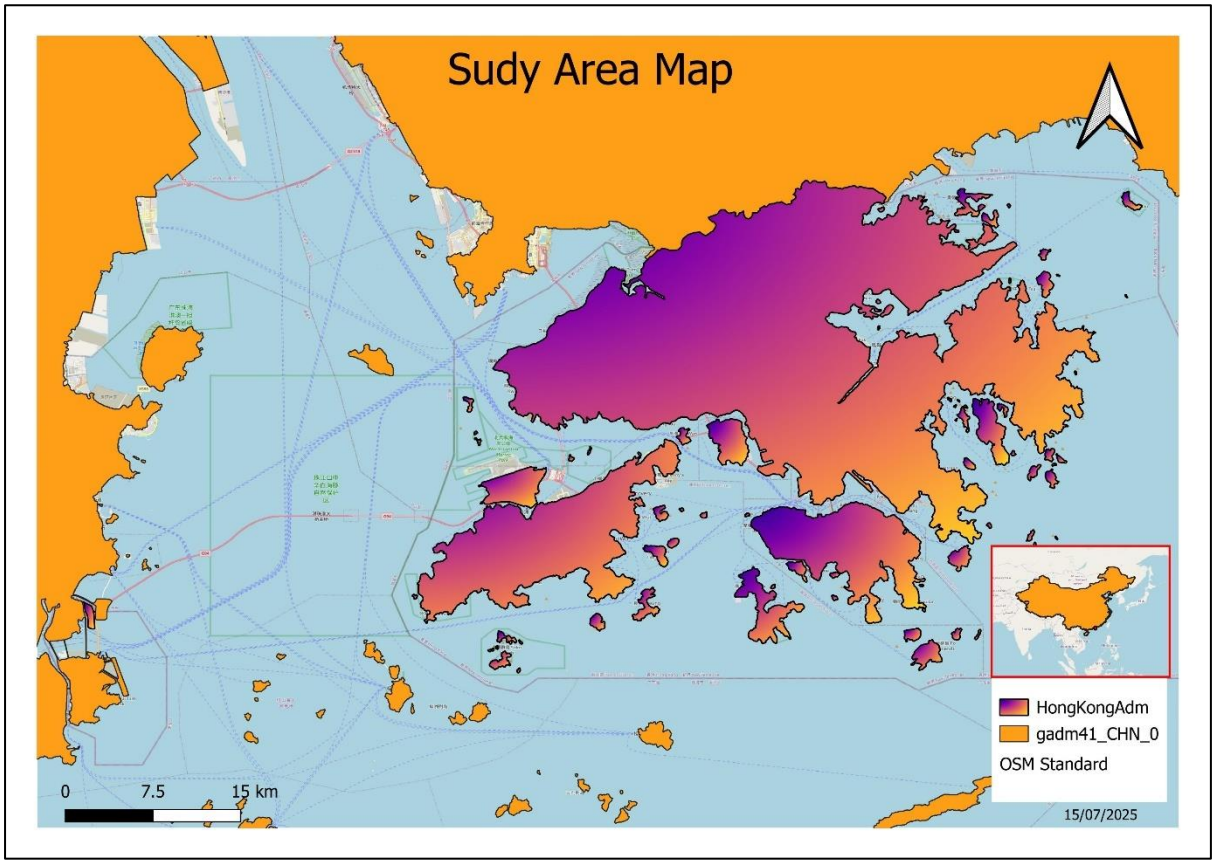


Figure 1 Hong Kong Administrative Boundary

2. Hydrological Risk Analysis

Using the DEM, perform terrain-based flood risk modelling:

- Derive slope, flow accumulation, and contributing zones
- Identify low-slope areas (<5°) that may be prone to surface water accumulation
- Generate a flood-prone zone polygon layer
- Compute exposure statistics for roads, hospitals, and utilities within these zones

Methodology

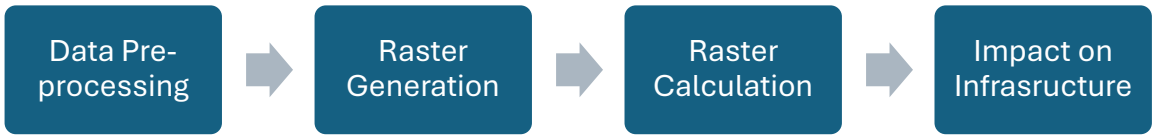
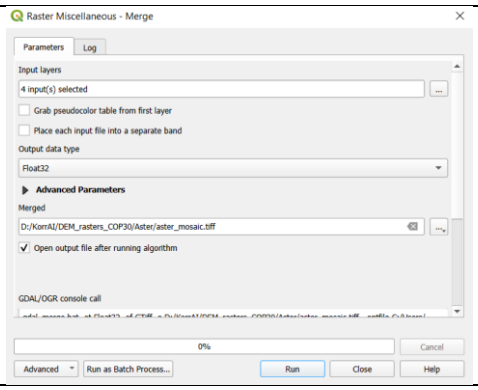
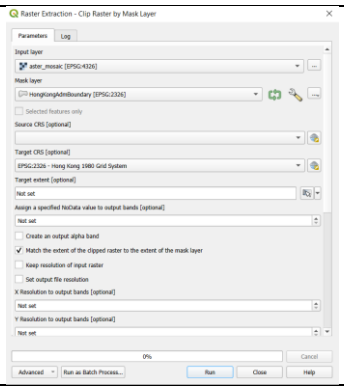


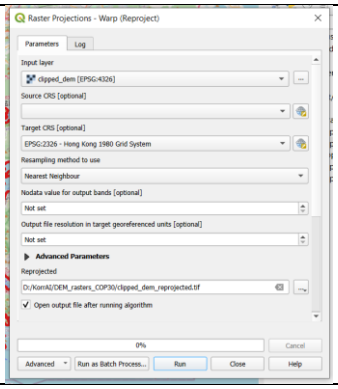
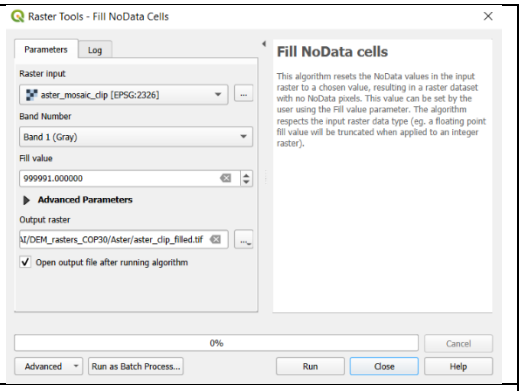
Figure 2 Methodology for Hydrological Risk Analysis

1. Data Preprocessing

a) Download ASTER DEM dataset tiles from the above links and open them in QGIS.
b) Mosaic ASTER DEM Tiles
c) Clip ASTER DEM mosaicked raster.
d) Reproject clipped DEM data to the local Hong Kong Projection – Hong Kong 1980 Grid System (EPSG: 2326)
e) Fill the missing values in DEM layer using Fill No Data Cells Tool in QGIS.

Table 2 Screens of steps followed in step 1

	
Mosaicking DEM Tiles	Clipping Mosaicked Data

	
Reproject DEM	Fill No Data Values

2. Data Generation

(Check the screenshots in below table for exact parameters)

a) Slope Raster Generation: Go to Raster -> Analysis -> Slope and fill the required parameters as shown in screen shot below.

b) Flow Accumulation Raster Generation:

   a. Install PC Raster Plugin in QGIS - https://www.youtube.com/watch?v=2pEr5RoGAXc

   b. Convert DEM data to PC Raster Format.

   c. Create Drain Direction Map using ldd create tool.

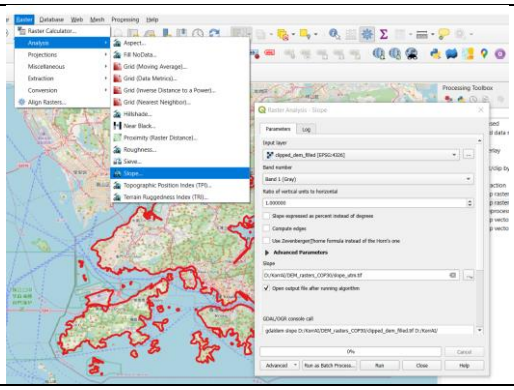
   d. Create Flow Accumulation Raster using accuflux tool in PC Raster.

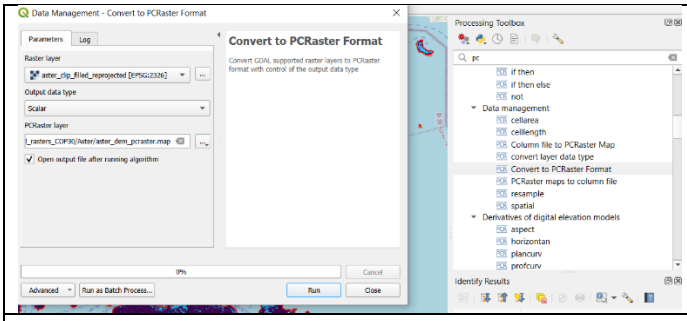
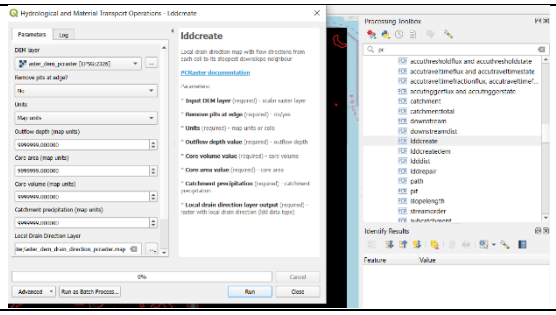
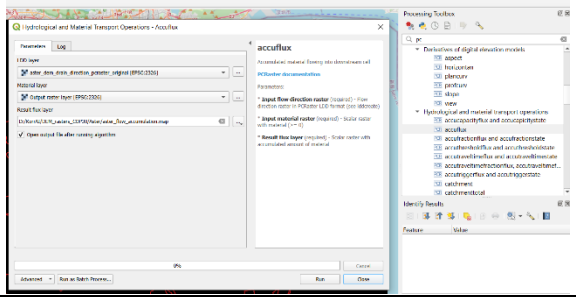
c) Generate Contours using DEM data.

d) Generate Hillshade raster using DEM data

e) Apply right symbology on the generated rasters.

Table 3 Screens of steps followed in step 2

	Generate slope raster

	Convert DEM data to PC Raster Format
	Create Drain Direction Map using ldd create tool
	Create Flow Accumulation Raster using accuflux tool in PC Raster

Maps generated in step 2

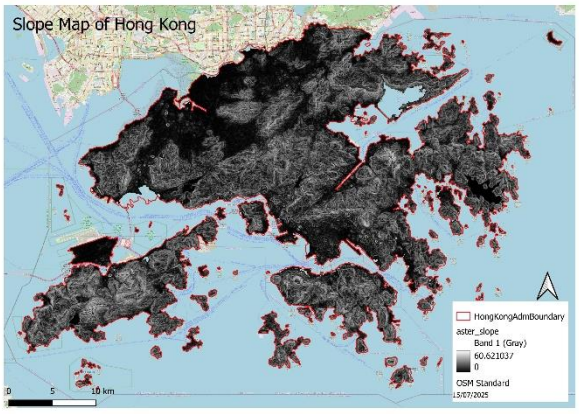


Figure 3 Slope Map (Source: Author)

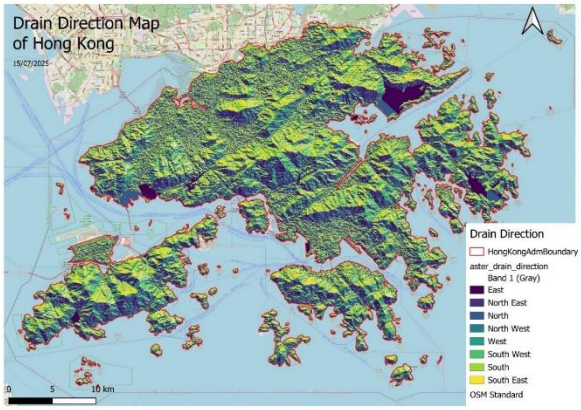


Figure 4 Drain Direction Map (Source: Author)

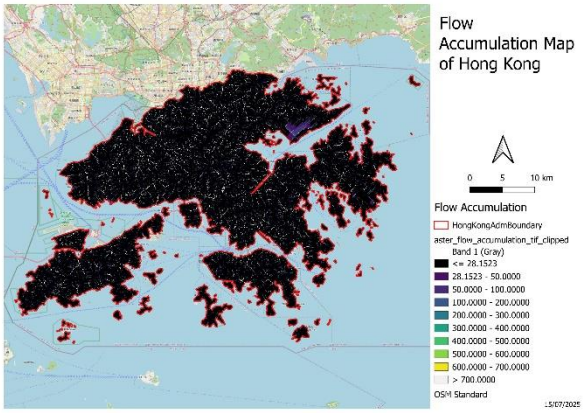


Figure 5 Flow Accumulation Map

3. Raster Calculations

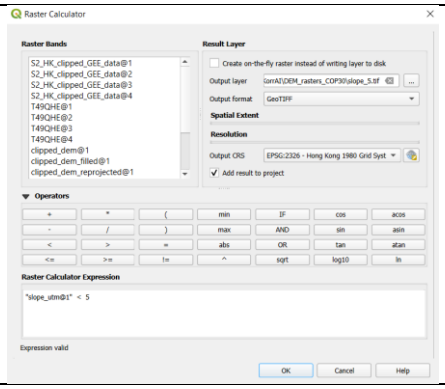
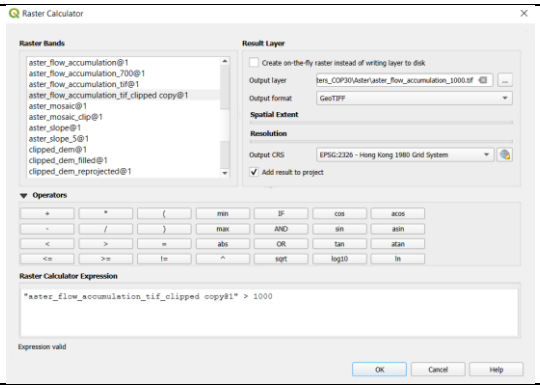
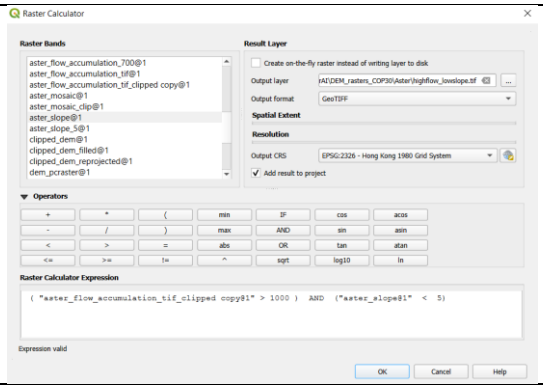
a. Low Slope Regions:
    i. Go to Raster -> Raster Calculator
    ii. Fill in the expression – “slope_utm@1” < 5. This will generate raster showing regions with slope less than 5 degrees. Score of 1 represents areas where slope is less than 5 degrees.

b. High Flow Accumulation Regions:
    i. Go to Raster -> Raster Calculator
    ii. Fill in the expression – “aster_flow_accumulation_tif_clipped@1” > 1000. This will generate raster showing regions with accumulation higher than 1000 raster cells. Score of 1 represents areas where accumulation is higher than 1000 raster cells.

c. Flood-Prone Zones - Low Slope and High Accumulation Regions
    i. Go to Raster -> Raster Calculator
    ii. Fill in the expression – (“aster_flow_accumulation_tif_clipped@1” > 1000) AND (“slope_utm@1” < 5). This will generate raster showing regions with accumulation higher than 1000 raster cells

and slope less than 5 degrees. These are the flood prone regions where in flow is high but out flow is low.

Table 4 Screens of steps followed in step 3

	
Calculating low slope areas in QGIS	Calculating high flow accumulation areas in QGIS
	
Calculating flood prone zones in QGIS	

Maps generated in Step 3 -

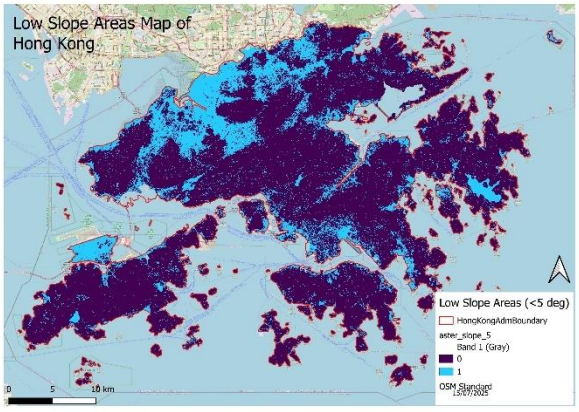


Figure 6 Low Slope Areas Map

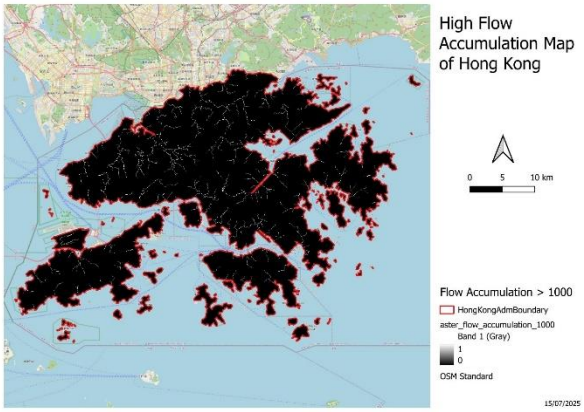


Figure 7 High Flow Accumulation Map

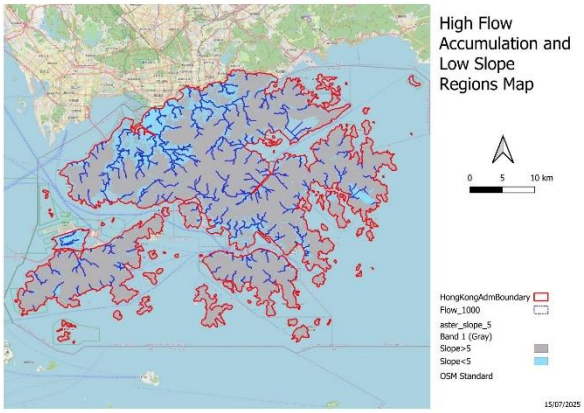


Figure 8 High Flow Accumulation Raster Overlaid Over Low Slope Regions

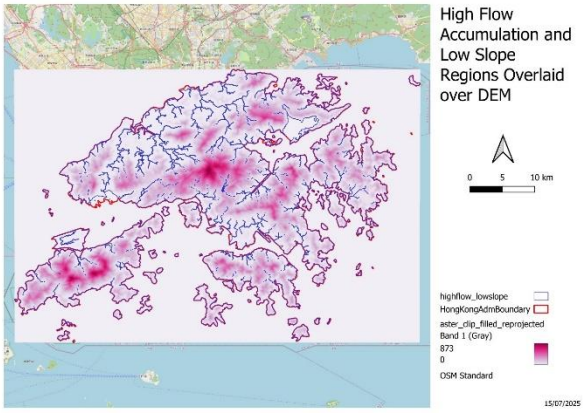
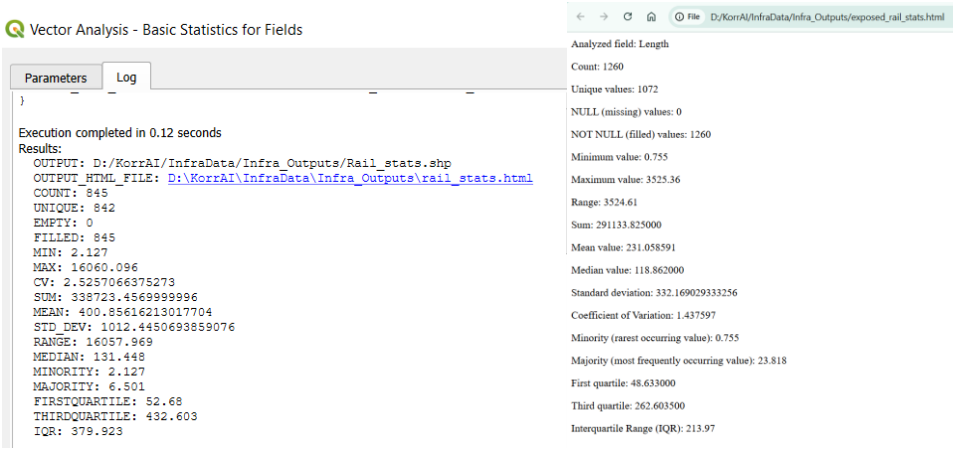


Figure 9 High Flow Low Slope Raster Generated Using Raster Calculator Visualized Over DEM Data

4. Impacted Infrastructure Mapping

d. Polygonise High Flow Low Slope Raster to create vector layer. The vector layer generated is in the form of streams. We need to create buffers around these stream vectors to get impacted regions.

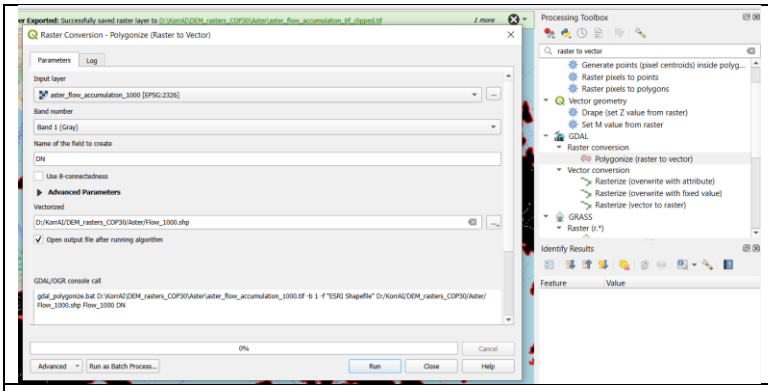
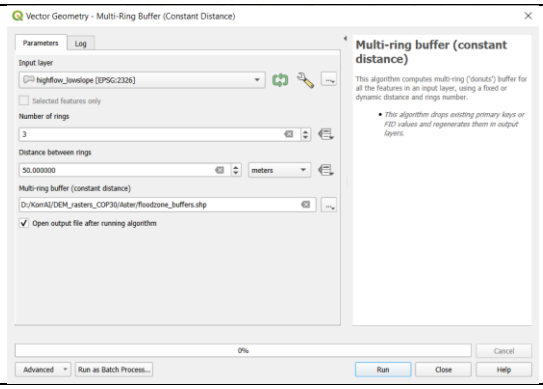
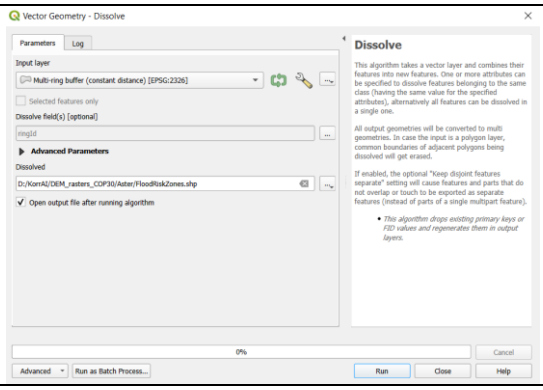
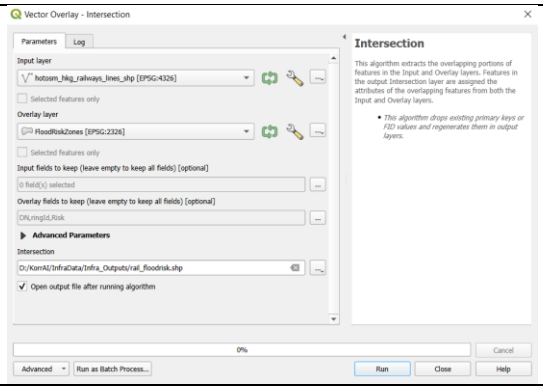
e. Create 3 multi-ring buffers of 50m distance each. This distance can vary as per analysis. First 50m is high risk zone, next 50m distance is medium risk zone and next 50 m is low risk zone.
    i. High risk (inner buffer) - Directly on flood pathways → high inundation & fast impact
    ii. Medium risk - Proximity to overflow zones; potential for shallow flooding
    iii. Low risk - Edge influence: surface ponding, poor drainage, or backup flooding
f. Dissolve these buffers to have a continuous polygon vector of flood risk zones.
g. Extract exposed railway lines:
    i. Open Hong Kong Railway Lines Shapefile.
    ii. Using geoprocessing, intersect these railway lines with flood risk zones to get highly likely to be impacted tracks during floods.
    iii. Create a new column in the output of intersection tool and calculate the length of railway lines likely to be impacted using field calculator.
    iv. Using Basic statistics for fields tool to get summary statistics of exposed railway lines. For current analysis, total railway line length = 338723.456 m. Exposed railway line length = 291133.825 m

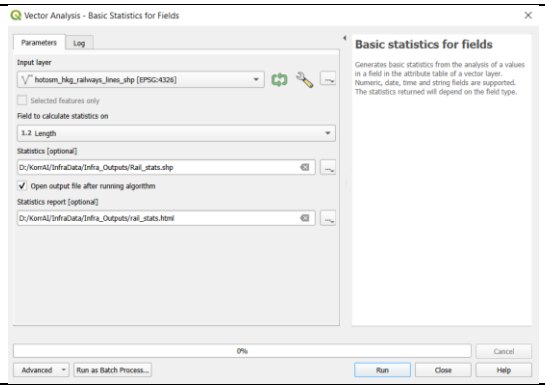
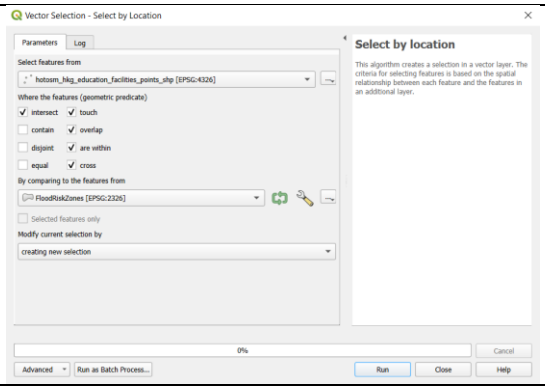


h. Extract exposed education facilities:
    i. Open education facilities point shapefile.
    ii. Using spatial query, select the education facilities falling within the flood risk zones. In this analysis, out of 192, 46 education facilities are exposed to flood risks. Out of 46 exposed education facilities, 18 are falling in high-risk flood zones (shown in maps below).
    iii. Follow the same process to get highly likely to be impacted health facilities during floods.

For this assessment, we have considered railway lines, education facilities and health facilities. Similarly, other likely to be impacted infrastructure analysis can also be carried out.

Table 5 Screens of steps followed in step 4

	Convert raster to vector
	Create multi-ring buffers
	Dissolve buffers using ring id.
	Extract exposed railway lines using intersection.

	Calculate summary statistics of exposed railway lines.
	Extract exposed education facilities

Maps generated in step 4 -

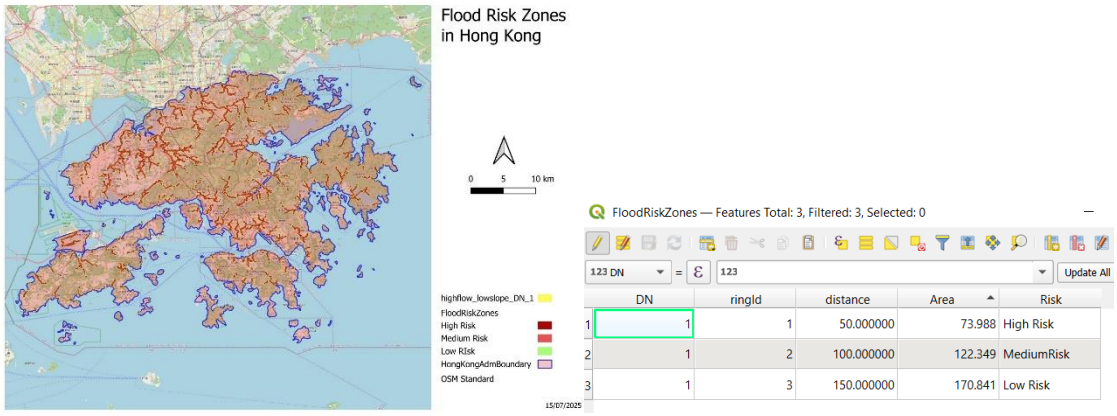


Figure 10 Flood Risk Zones with Area (km2) Under Each Risk Zone

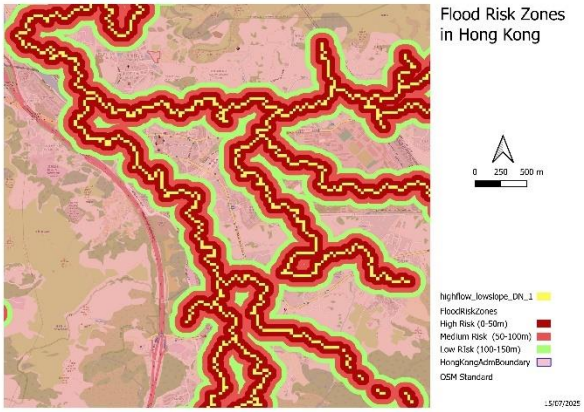


Figure 11 Flood Risk Zones (Zoomed In)

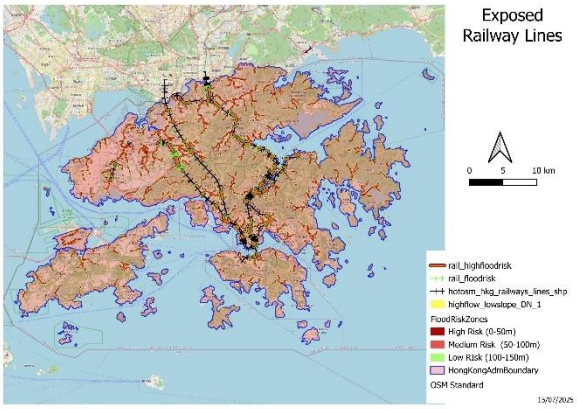


Figure 12 Exposed Railway Tracks

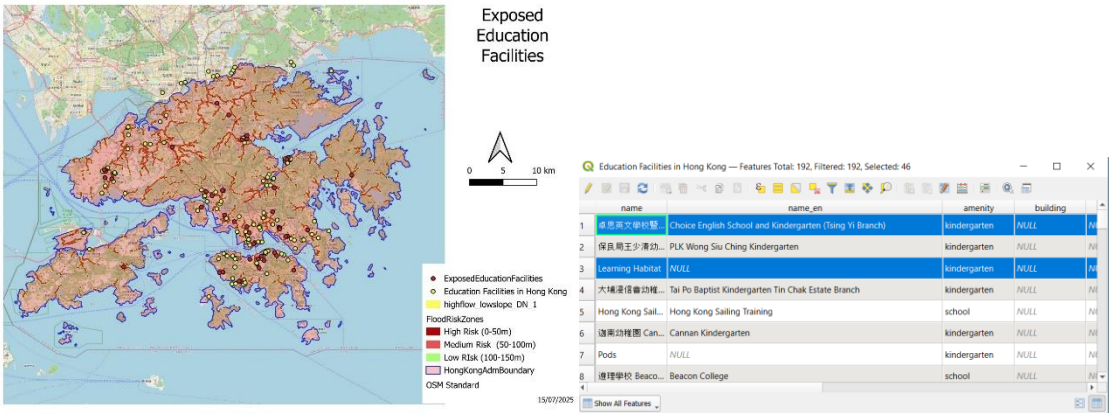


Figure 13 Exposed Education Facilities in Flood Risk Zones with Count shown in Attribute Table

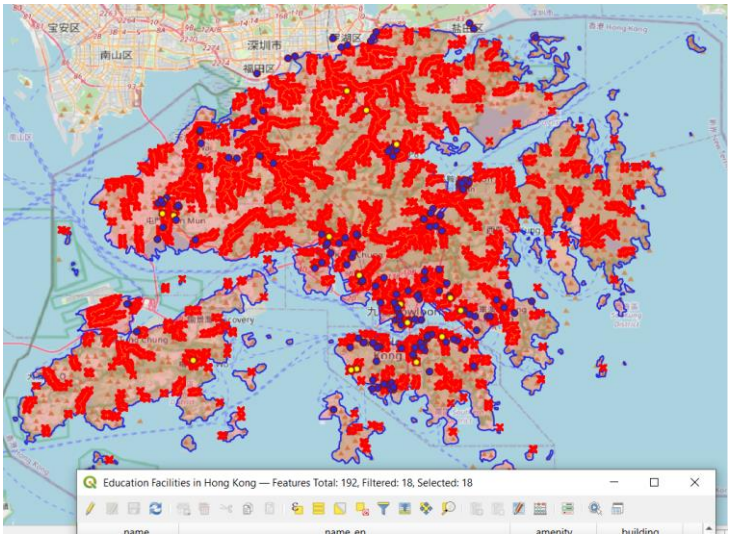


Figure 14 Exposed Education Facilities in High Flood Risk Zones

3. Historical Flood Exposure Assessment

Using at least two historical floodplain datasets (e.g., 2010, 2015, 2020):

- **Merge them to derive a flood frequency index raster**
- **Identify persistent flooding zones**
- **Intersect these zones with infrastructure layers**
- **Summarize exposure by type and zone**
- **Submit a tabulated summary in CSV format**

Methodology

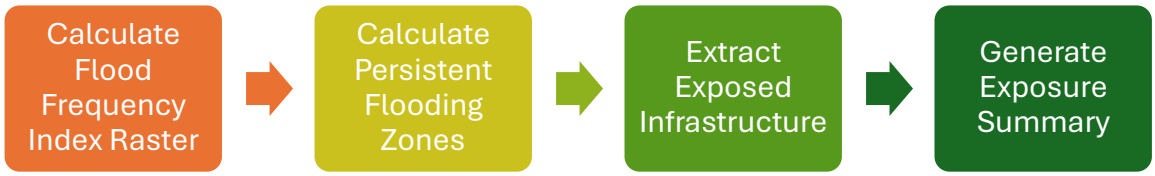


Figure 15 Methodology for Historical Flood Exposure Assessment

1. Calculate flood frequency index raster

a. Download historical flood plain datasets from Global Flood Dataset.
b. For this analysis, we have considered Hong Kong flood datasets of 2006, 2007 and 2014.

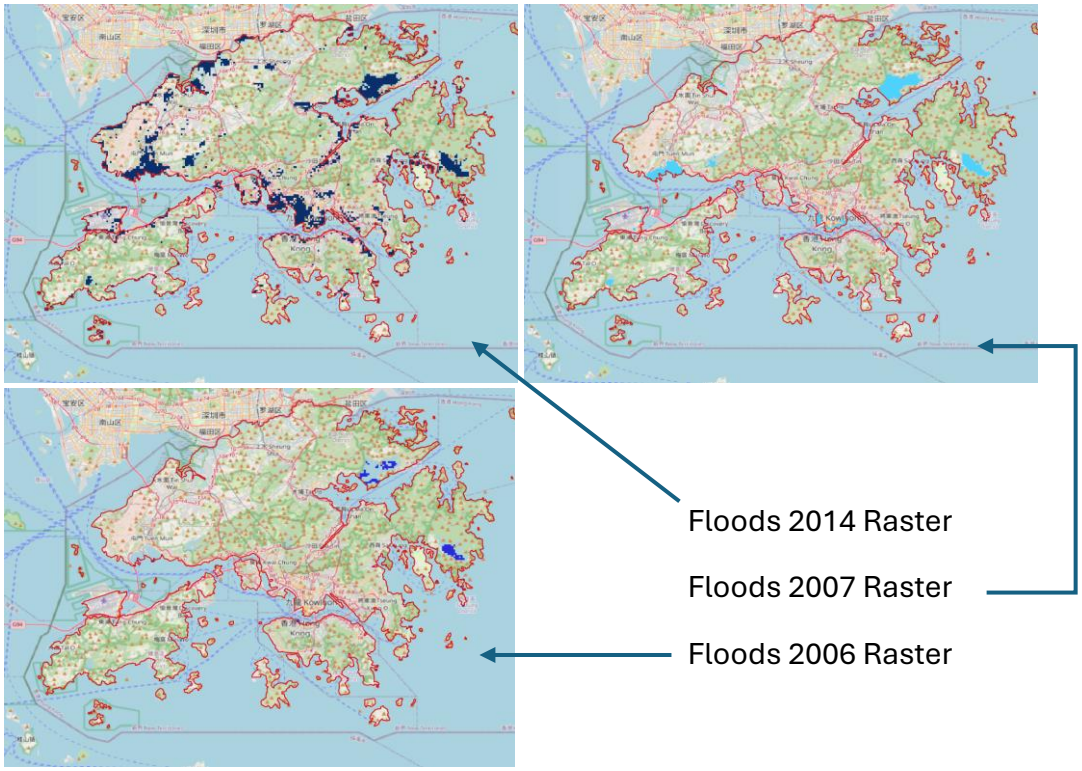
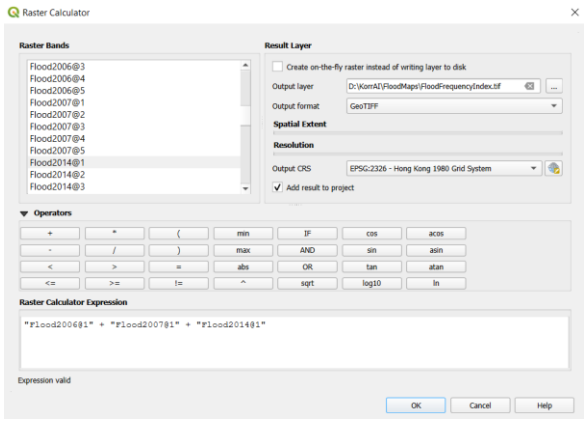


Figure 16 Historical Flood Dataset of Hong Kong

c. Add the three rasters using raster calculator to get the flood frequency of the regions. Flood Frequency Index values show the number of times a particular area faced floods. In the output, score of '1' means that the area was impacted by floods only once out of the three time periods. Score of '2' means that the area was impacted by floods twice out of the three time periods. Likewise for score of '3'. This gives a **flood frequency index**:

- 0 = never flooded
- 1 = flooded once
- 2 = flooded twice
- 3 = flooded in all events

Table 6 Screens of steps followed in step 1

	Flood Frequency Index Raster Calculation

Maps generated in step 1

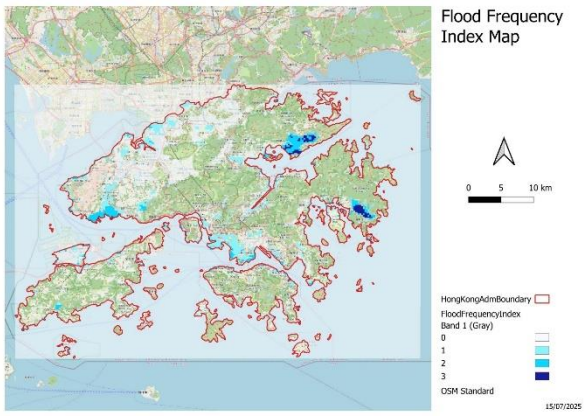


Figure 17 Flood Frequency Index Map

2. Persistent Flooding zones

a. Using raster calculator, extract pixels with flood frequency index values >= 2. These are the regions with persistent flooding.
b. Polygonise Persistent Flooding raster to create a vector layer.

Maps generated in step 2

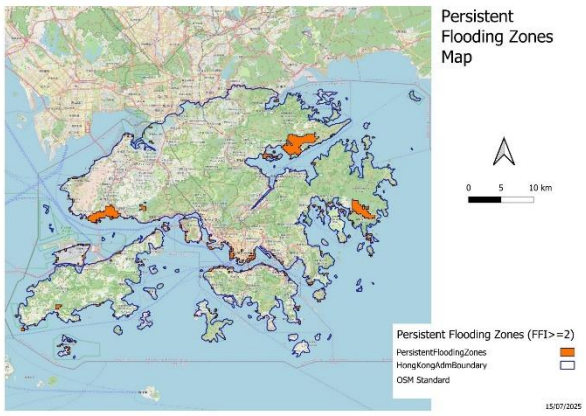


Figure 18 Persistent Flooding Zones

3. Extract Exposed Infrastructure layers

a. Extract exposed railway lines:
    i. Open Hong Kong Railway Lines Shapefile.
    ii. Using geoprocessing, intersect these railway lines with persistent flooding zones to get railway lines under persistent flooding zones.
    iii. Create a new column in the output of intersection tool and calculate the length of railway lines impacted using field calculator.
    iv. Using Basic statistics for fields tool to get summary statistics of exposed railway lines. For current analysis, total railway line length = 338723.456 m. Exposed railway line length = 4502 m

```
COUNT: 32
UNIQUE: 32
EMPTY: 0
FILLED: 32
MIN: 2.332
MAX: 317.39
CV: 0.4968126361296306
SUM: 4502.095
MEAN: 140.69046875
STD_DEV: 69.89680265800092
RANGE: 315.058
MEDIAN: 124.3945
MINORITY: 2.332
MAJORITY: 2.332
FIRSTQUARTILE: 99.19200000000001
THIRDQUARTILE: 181.4845
IQR: 82.29249999999999
```

Figure 19 Summary stats of exposed railway lines

b. Extract exposed education facilities:
    i. Open education facilities point shapefile.
    ii. Using spatial query, select the education facilities falling within the persistent flood risk zones. In this analysis, out of 192, 2 education facilities are exposed to persistent flood risks.
    iii. Follow the same process to get highly likely to be impacted health facilities during floods.

Maps generated in step 3.

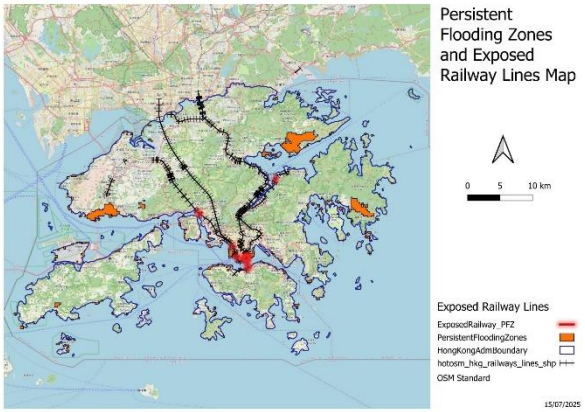


Figure 20 Exposed Railway Lines under Persistent Flooding Zones

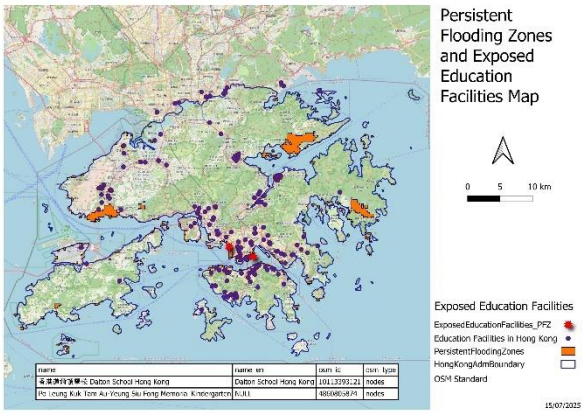


Figure 21 Exposed Education Facilities under Persistent Flooding Zones

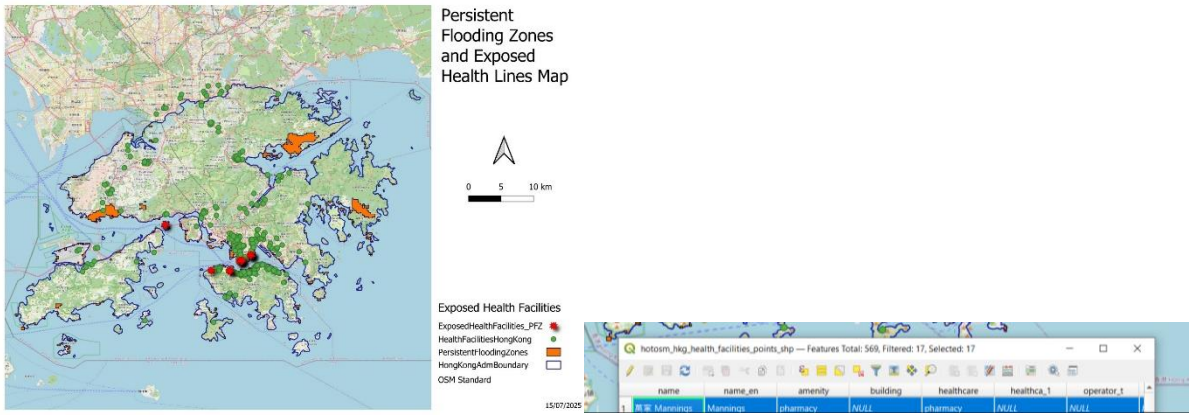


Figure 22 Exposed Health Facilities under Persistent Flooding Zones with Count in Attribute Table

4. Exposure Summary

Table 7 Exposure summary by type and zone

Infrastructure Type	Flood Frequency Zone	Impacted Count	Total Count	Impacted Length (km)	Total Length	% Impacted
Railway Lines	>=2			4.502 km	338.723 km	1.33%
Health Facility	>=2	17	569			2.99%
Education Facility	>=2	2	192			1.04 %

Analysis of persistent flood zones in Hong Kong (flood frequency ≥2) revealed that **1.33% of railway line length**, **2.99% of health facilities**, and **1.04% of educational facilities** are exposed to recurrent flooding. These exposed assets may require prioritization in flood resilience planning.

4. Land Use and Land Cover (LULC) Classification from Sentinel-2

Generate a five-class LULC raster using Sentinel-2 imagery with the following classes:

- **Urban**
- **Forest**
- **Water**
- **Agriculture**
- **Barren/Other**

Methodology

1. Acquiring Sentinel 2 Data

a. Download data from data portal (Copernicus or GEE).
b. Layer Stack bands using Merge tool in QGIS (not required if data has been acquired from GEE).
c. Create Mosaic using Merge Tool in QGIS (if area is covered within multiple tiles).
d. Clip raster using mask layer to get data of the desired region.
e. Reproject raster to the Hong Kong Local Projection 1980 to maintain consistency within datasets.
f. Change band combination to False Colour Composite using Symbology.

Map generated in step 1

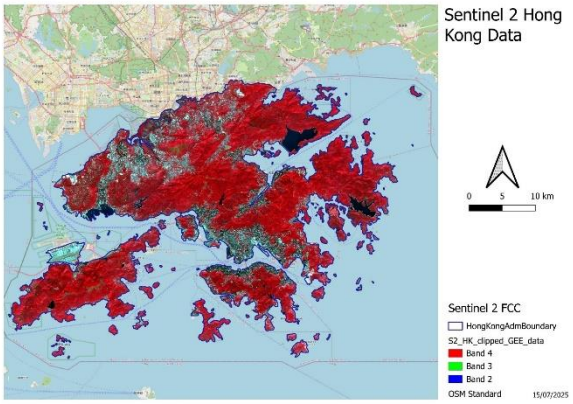
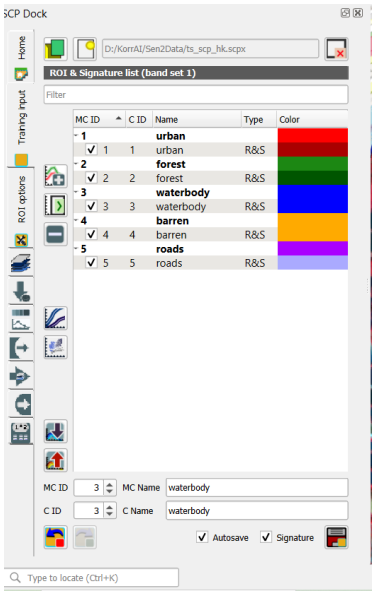


Figure 23 Acquired Sentinel 2 Data of Hong Kong in False Colour Composite

2. Supervised Classification

a. Install Semi-Automatic Classification Plugin in QGIS.
b. Generate training samples using SCP plugin for each class to be extracted. For this analysis, classes considered are – urban, forest, waterbody, barren and roads.



c. Perform Supervised Classification.
d. Generate confusion matrix to check accuracy.

Map generated in step 2

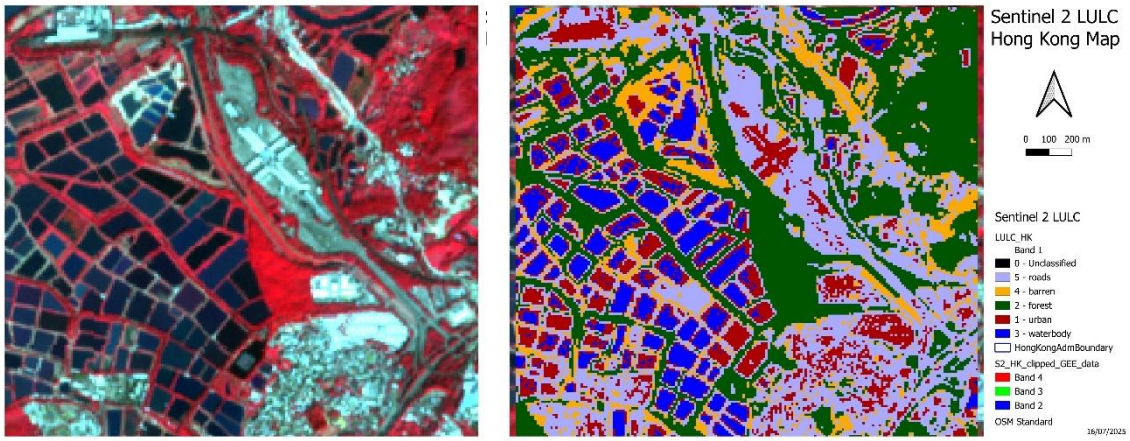


Figure 24 LULC Map of Hong Kong (Zoomed In)

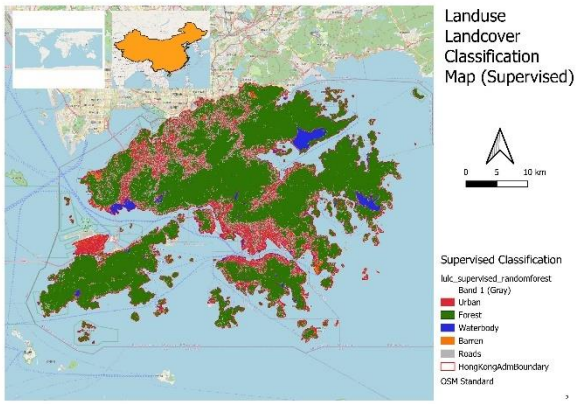
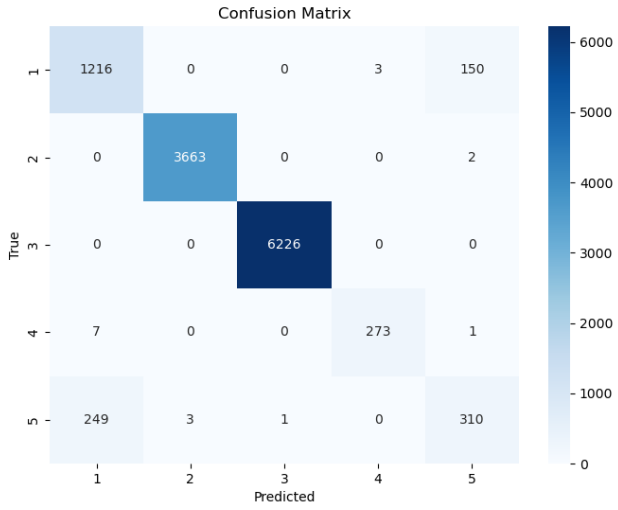


Figure 25 Supervised Classification Map

Classification Report:

	precision	recall	f1-score	support
1	0.83	0.89	0.86	1369
2	1.00	1.00	1.00	3665
3	1.00	1.00	1.00	6226
4	0.99	0.97	0.98	281
5	0.67	0.55	0.60	563
accuracy			0.97	12104
macro avg	0.90	0.88	0.89	12104
Weighted avg	0.96	0.97	0.96	12104



3. Unsupervised Classification

a. Perform unsupervised classification using KMeans algorithm in SCP Plugin. Number of classes = 5.

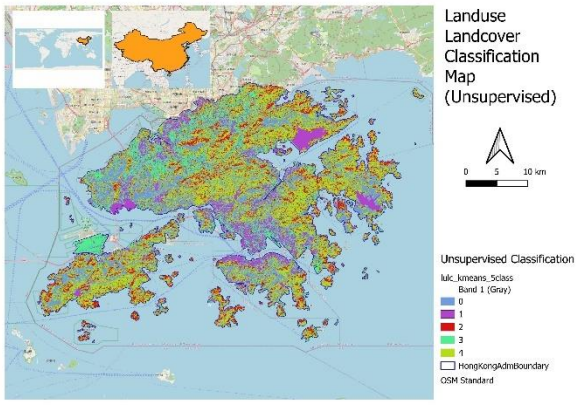


Figure 26 Unsupervised Classification Map of Hong Kong

4. Python Automation Pipeline

Develop a Python-based workflow that automates the following:

- **Data reprojection and clipping**
- **Hydrological modeling**
- **LULC classification**
- **Flood analysis and zonal statistics**

Your pipeline should accept input parameters and output organized folders.
Use libraries such as geopandas, rasterio, fiona, richdem, and numpy.
Bonus: Add a command-line interface (CLI) or a basic Streamlit-based front end for running the pipeline.

Script attached in folder.

5. Cartographic Output

Produce five professionally designed maps using QGIS:

1. **LULC Classification Map**
2. **Flood Frequency Index Map**
3. **Topographic Risk Zones Map**
4. **Infrastructure Exposure Map**
5. **Terrain Map (Hillshade + Contours)**

Each map must include:

- **Scalebar, legend, and north arrow**
- **Inset map for geographic context**
- **Proper symbology and labels**
- **Source attribution**

Required Maps Attached Below

1. LULC Classification Map

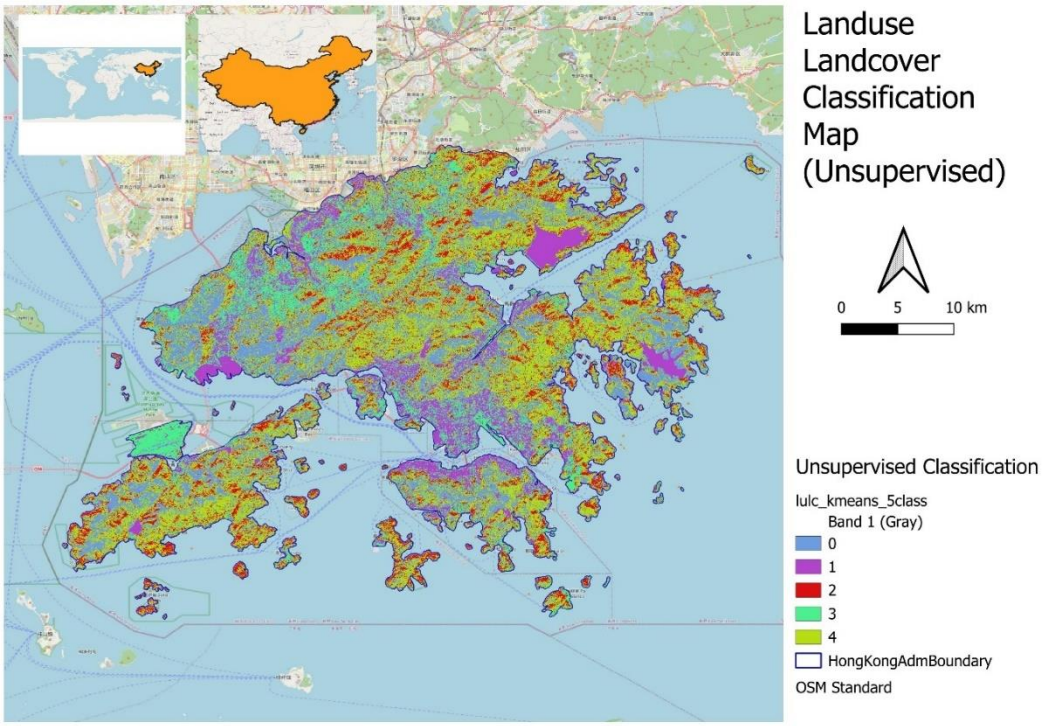


Figure 27 LULC Classification Map (Unsupervised) - Source: Author

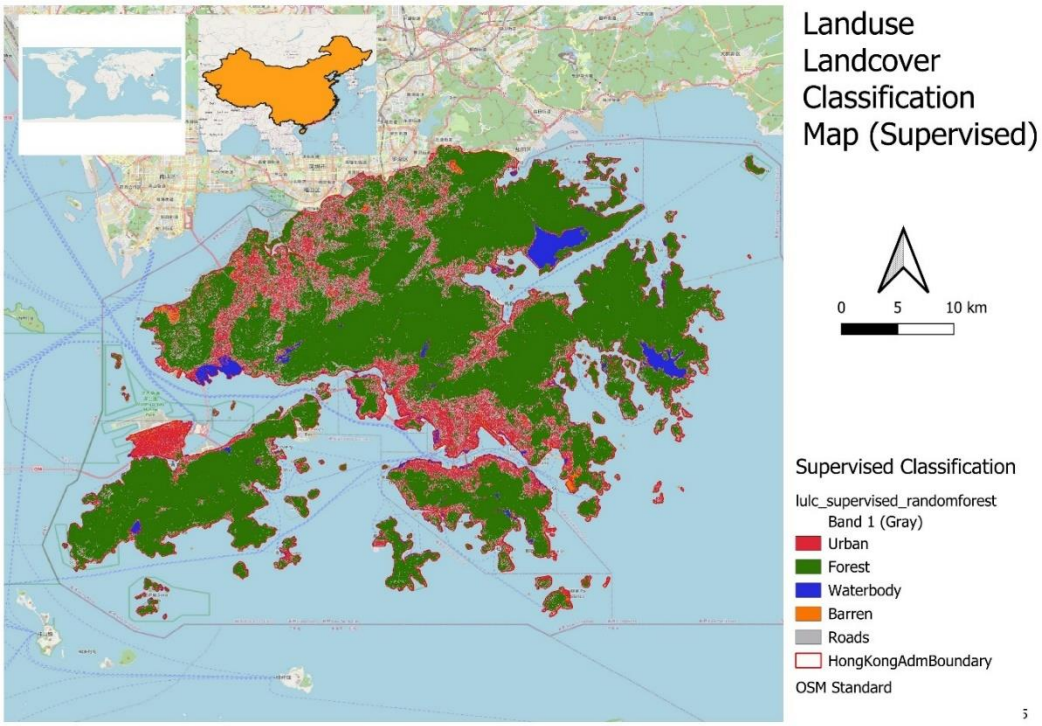


Figure 28 Supervised Classification

2. Flood Frequency Index Map

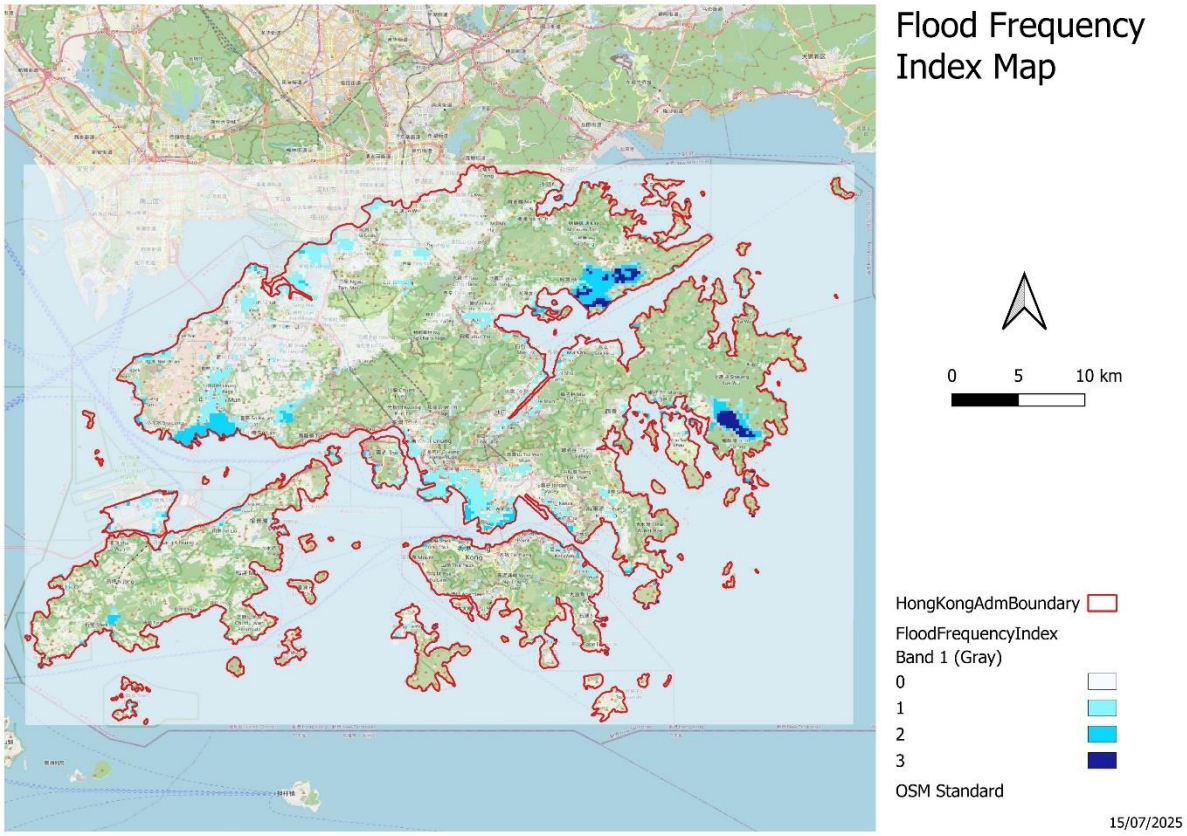
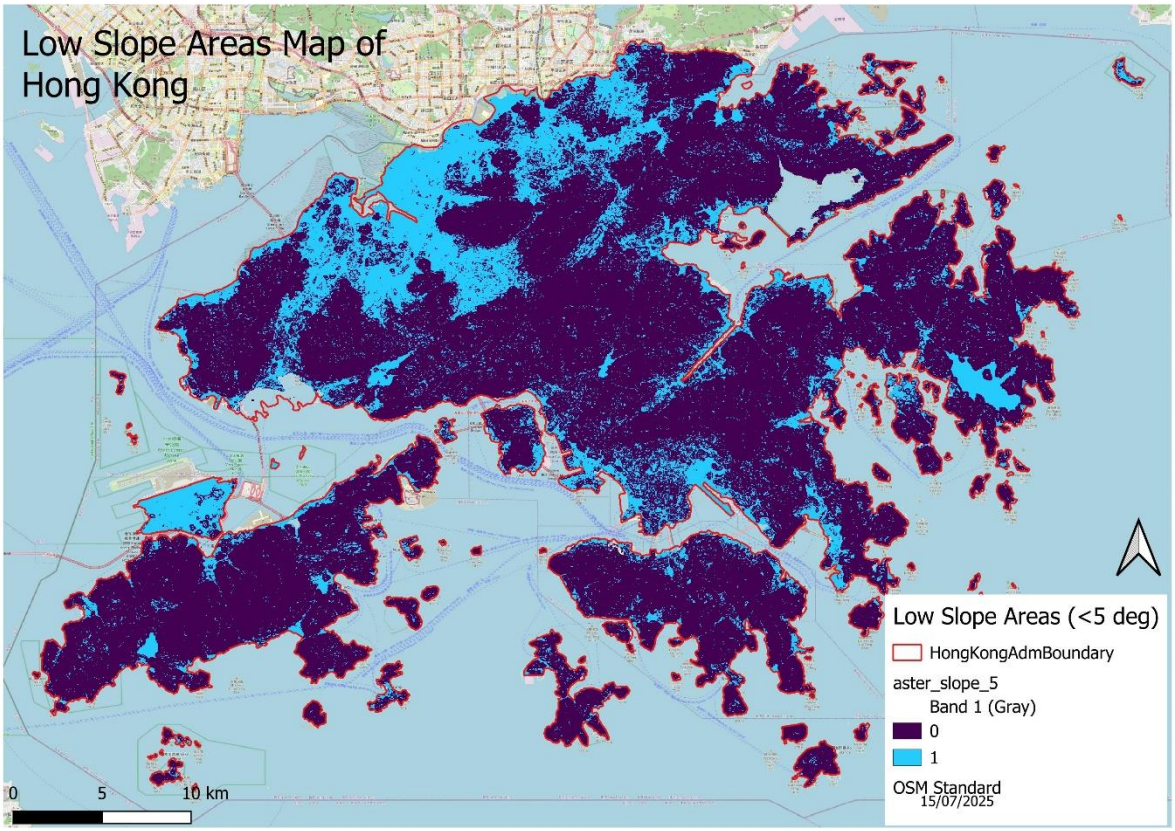
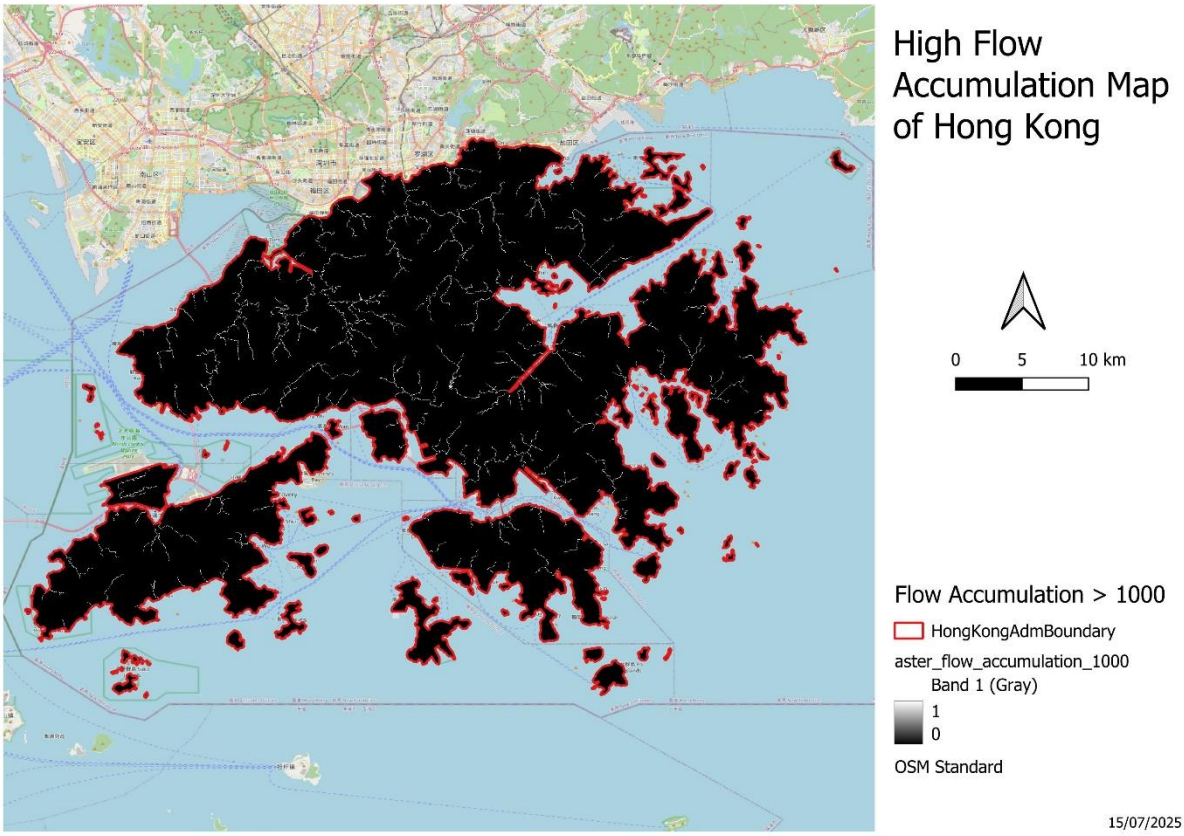
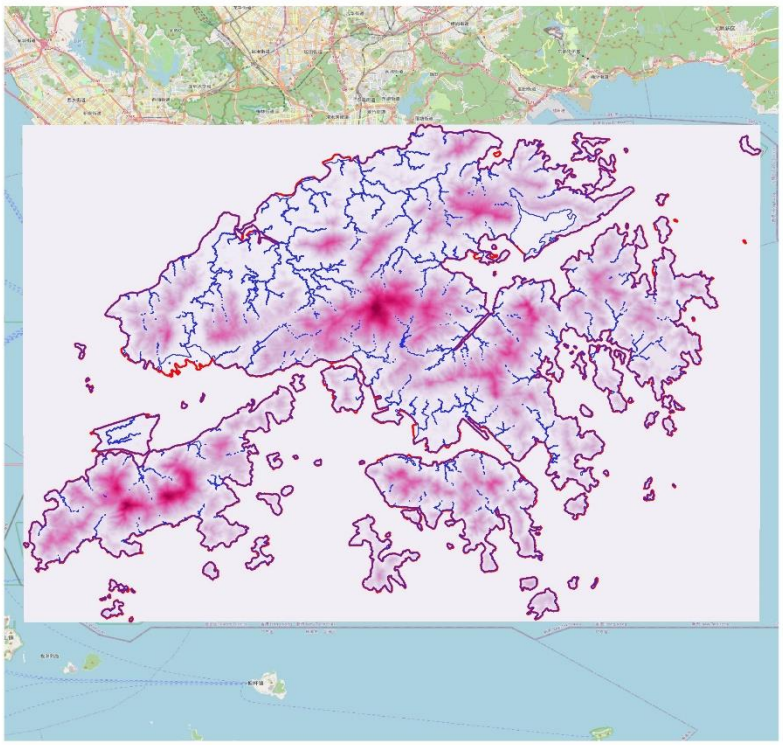


Figure 29 Flood Frequency Index Map

3. Topographic Risk Zones Map





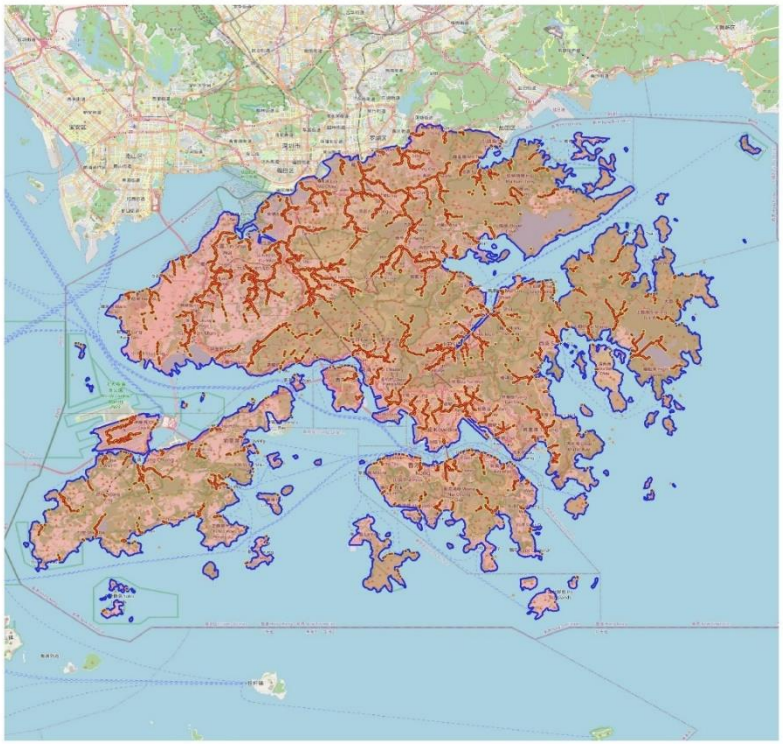


High Flow Accumulation and Low Slope Regions Overlaid over DEM

0 5 10 km

highflow_lowslope
HongKongAdmBoundary
aster_clip_filled_reprojected
Band 1 (Gray)
873
0
OSM Standard

15/07/2025

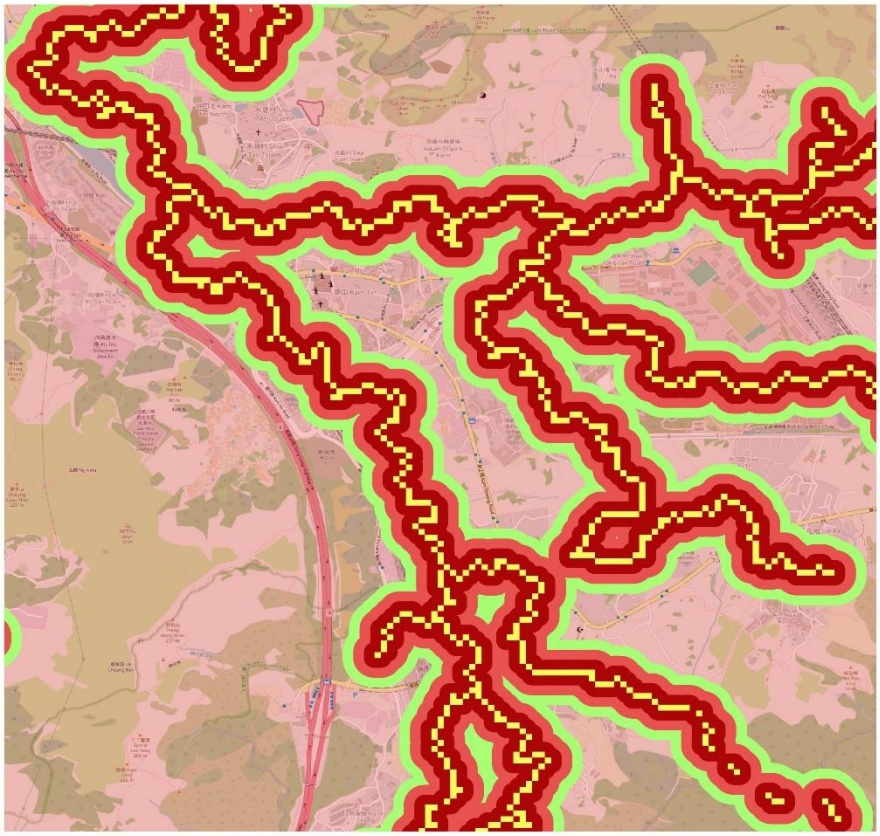


Flood Risk Zones in Hong Kong

0 5 10 km

highflow_lowslope_DN_1
FloodRiskZones
High Risk
Medium Risk
Low RIsk
HongKongAdmBoundary
OSM Standard

15/07/2025



Flood Risk Zones in Hong Kong

0 250 500 m

highflow_lowslope_DN_1

FloodRiskZones

High Risk (0-50m)

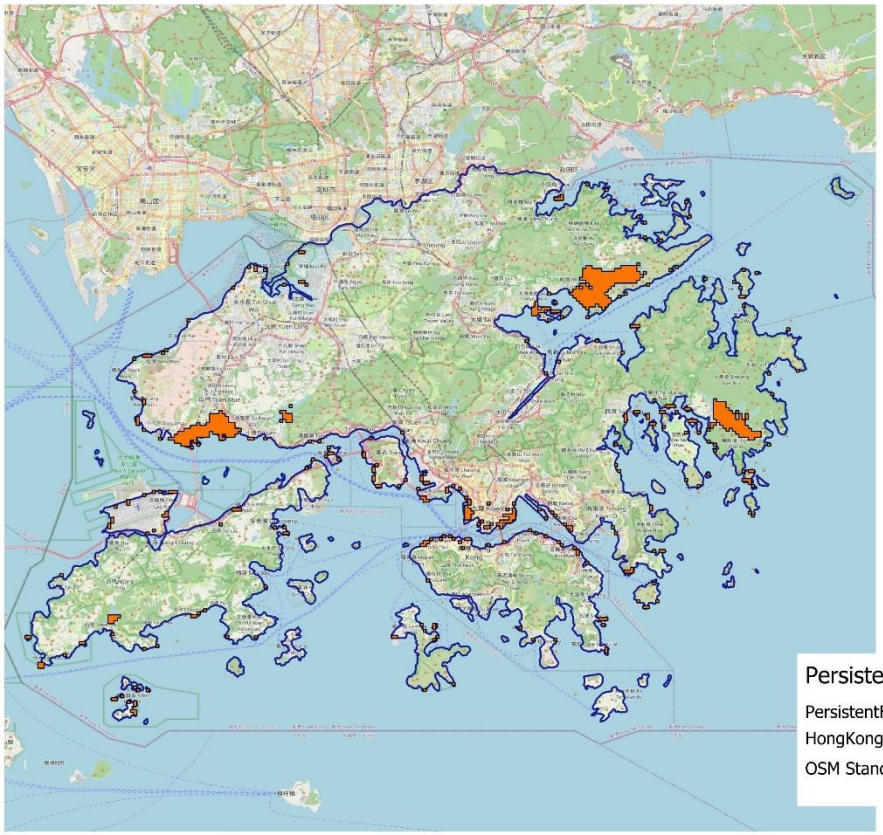
Medium Risk (50-100m)

Low RIsk (100-150m)

HongKongAdmBoundary

OSM Standard

15/07/2025



Persistent Flooding Zones Map

0 5 10 km

Persistent Flooding Zones (FFI>=2)

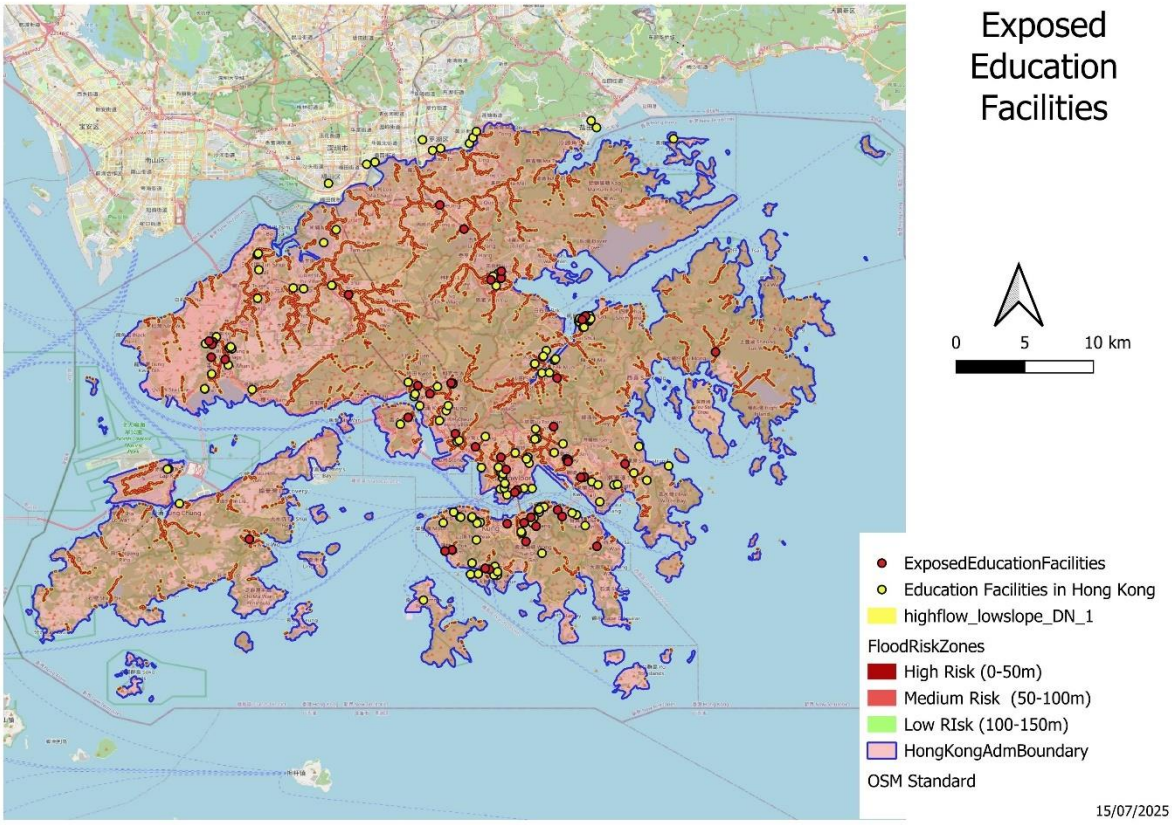
PersistentFloodingZones

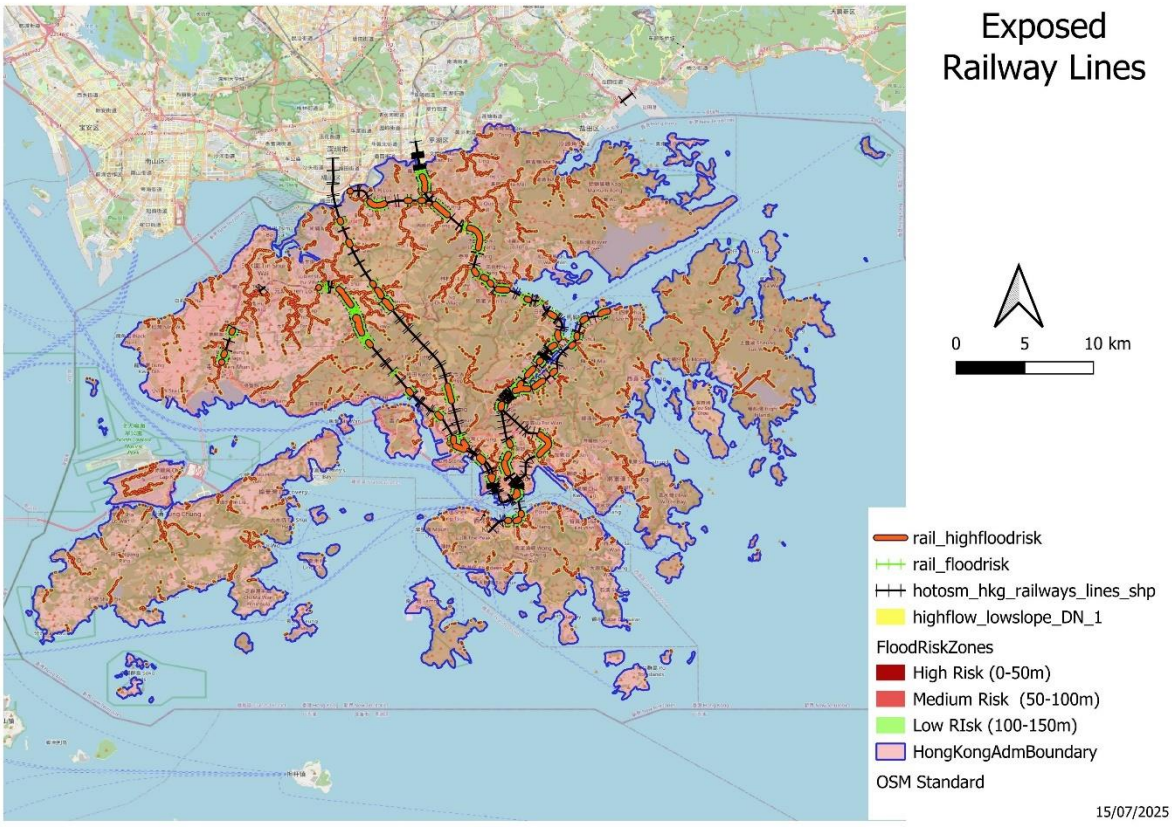
HongKongAdmBoundary

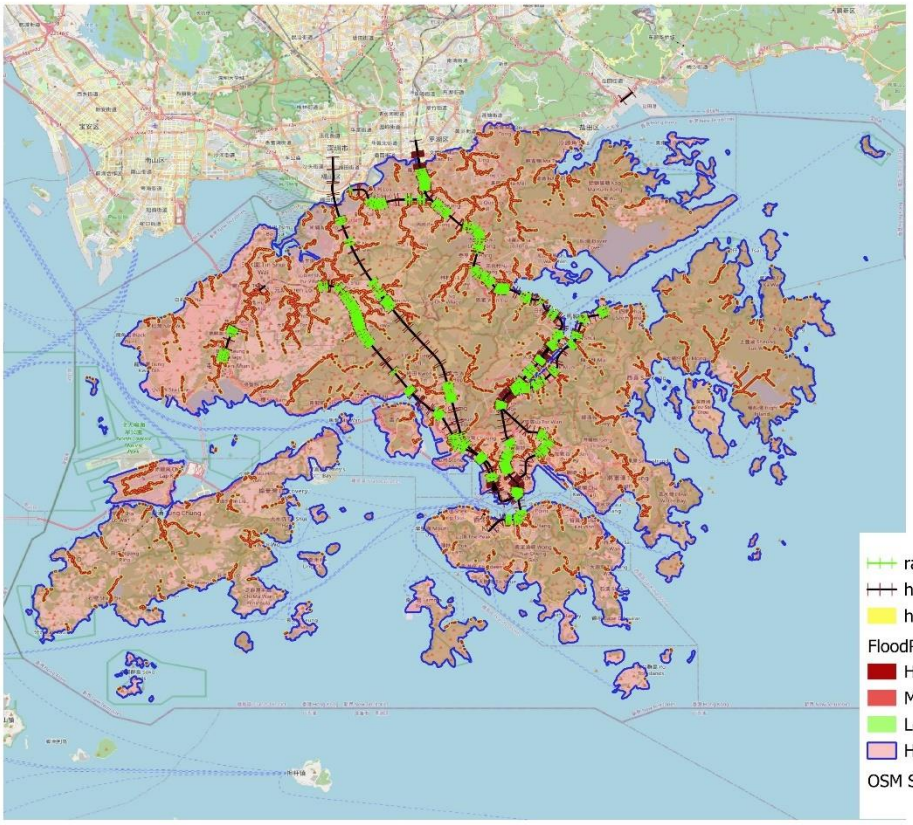
OSM Standard

15/07/2025

4. Infrastructure Exposure Map





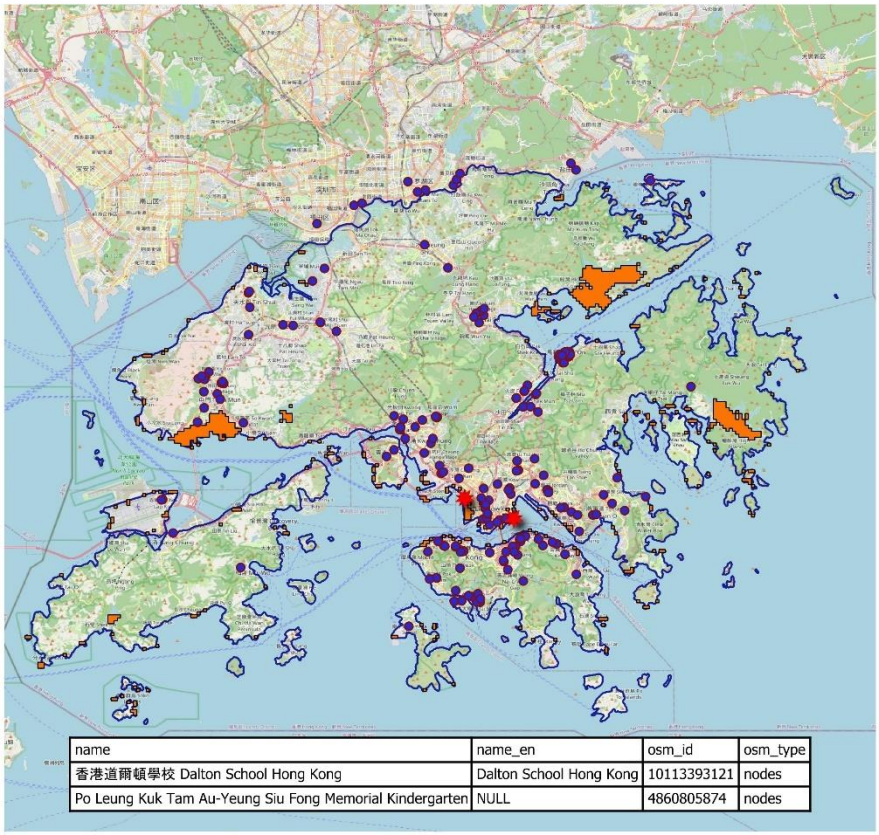


Exposed Railway Lines

0 5 10 km

rail_floodrisk
hotosm_hkg_railways_lines_shp
highflow_lowslope_DN_1
FloodRiskZones
High Risk (0-50m)
Medium Risk (50-100m)
Low RIsk (100-150m)
HongKongAdmBoundary
OSM Standard

15/07/2025



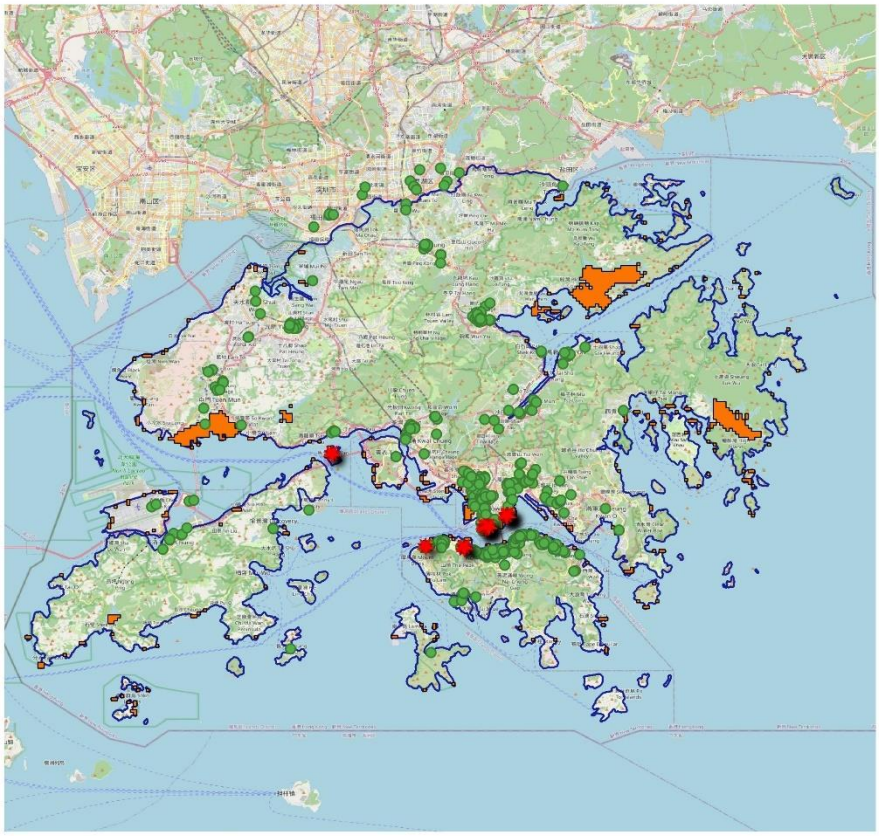
Persistent Flooding Zones and Exposed Education Facilities Map

0 5 10 km

Exposed Education Facilities
ExposedEducationFacilities_PFZ
Education Facilities in Hong Kong
PersistentFloodingZones
HongKongAdmBoundary
OSM Standard

15/07/2025

name	name_en	osm_id	osm_type
香港道爾頓學校 Dalton School Hong Kong	Dalton School Hong Kong	10113393121	nodes
Po Leung Kuk Tam Au-Yeung Siu Fong Memorial Kindergarten	NULL	4860805874	nodes



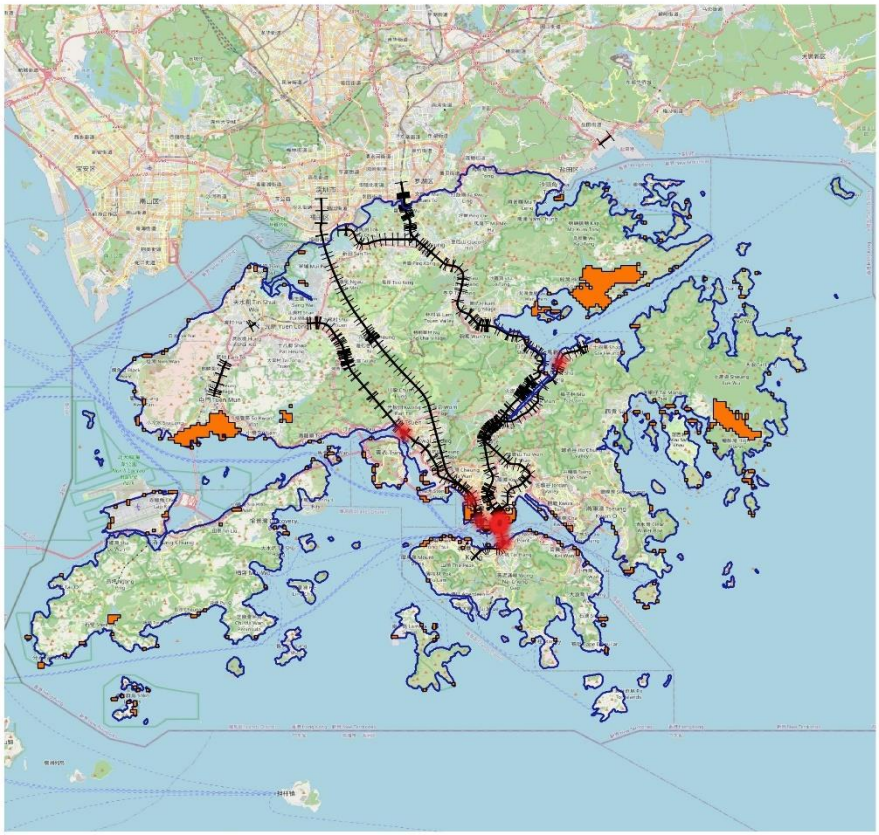
Persistent Flooding Zones and Exposed Health Lines Map

0 5 10 km

Exposed Health Facilities

ExposedHealthFacilities_PFZ
HealthFacilitiesHongKong
PersistentFloodingZones
HongKongAdmBoundary
OSM Standard

15/07/2025



Persistent Flooding Zones and Exposed Railway Lines Map

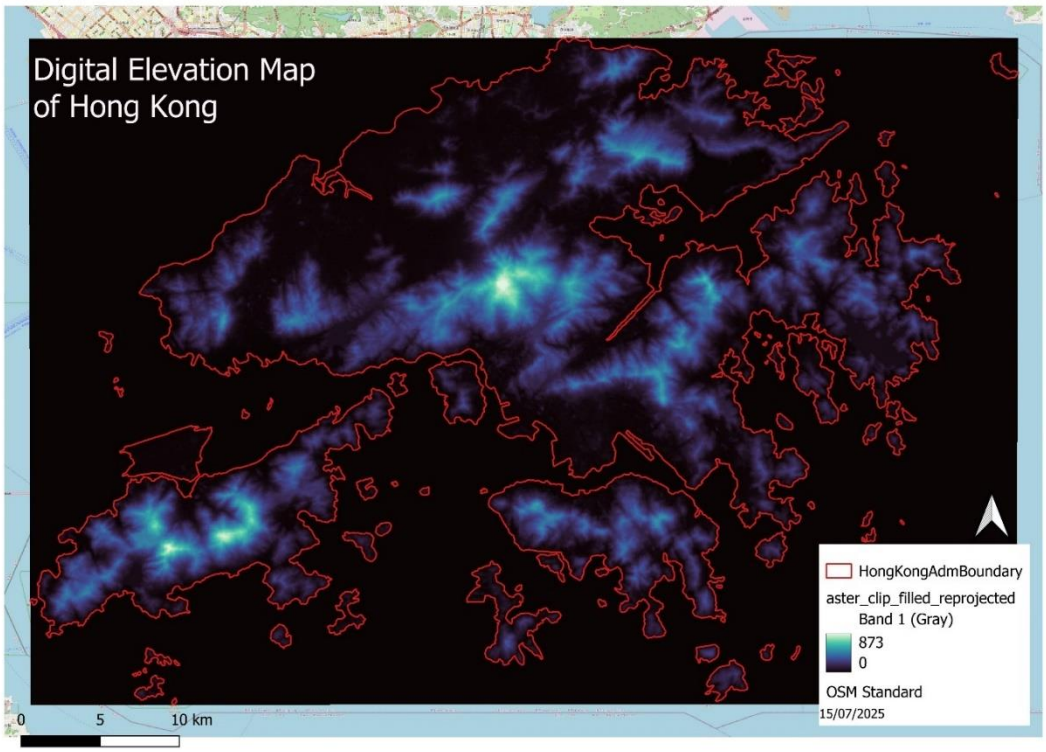
0 5 10 km

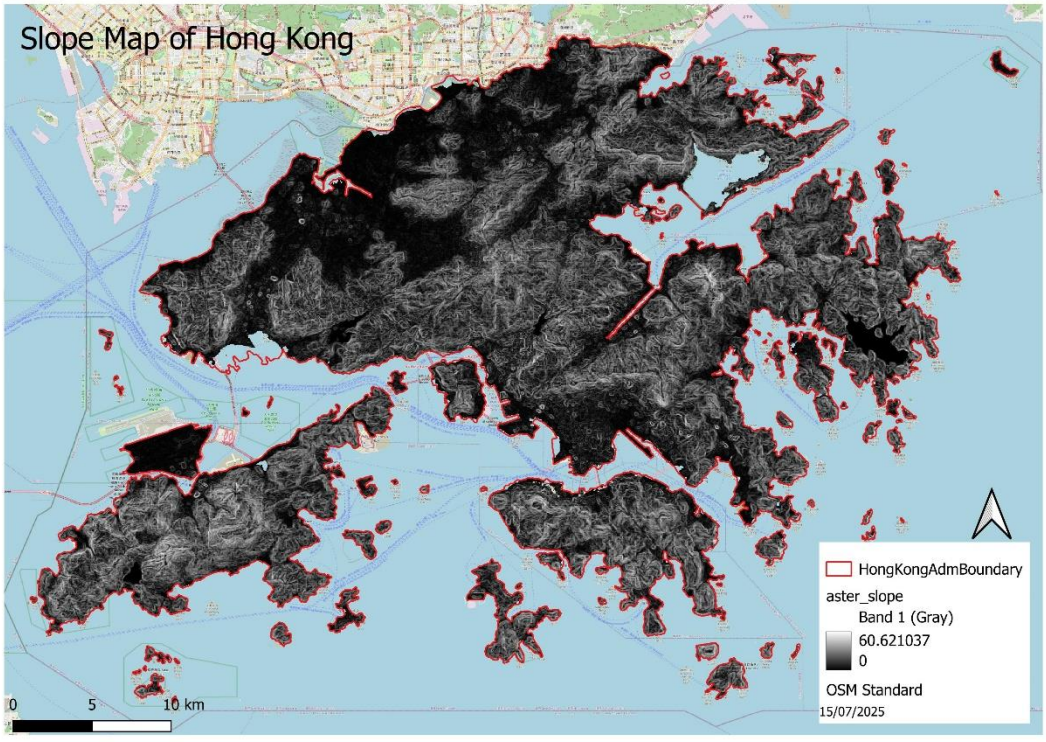
Exposed Railway Lines

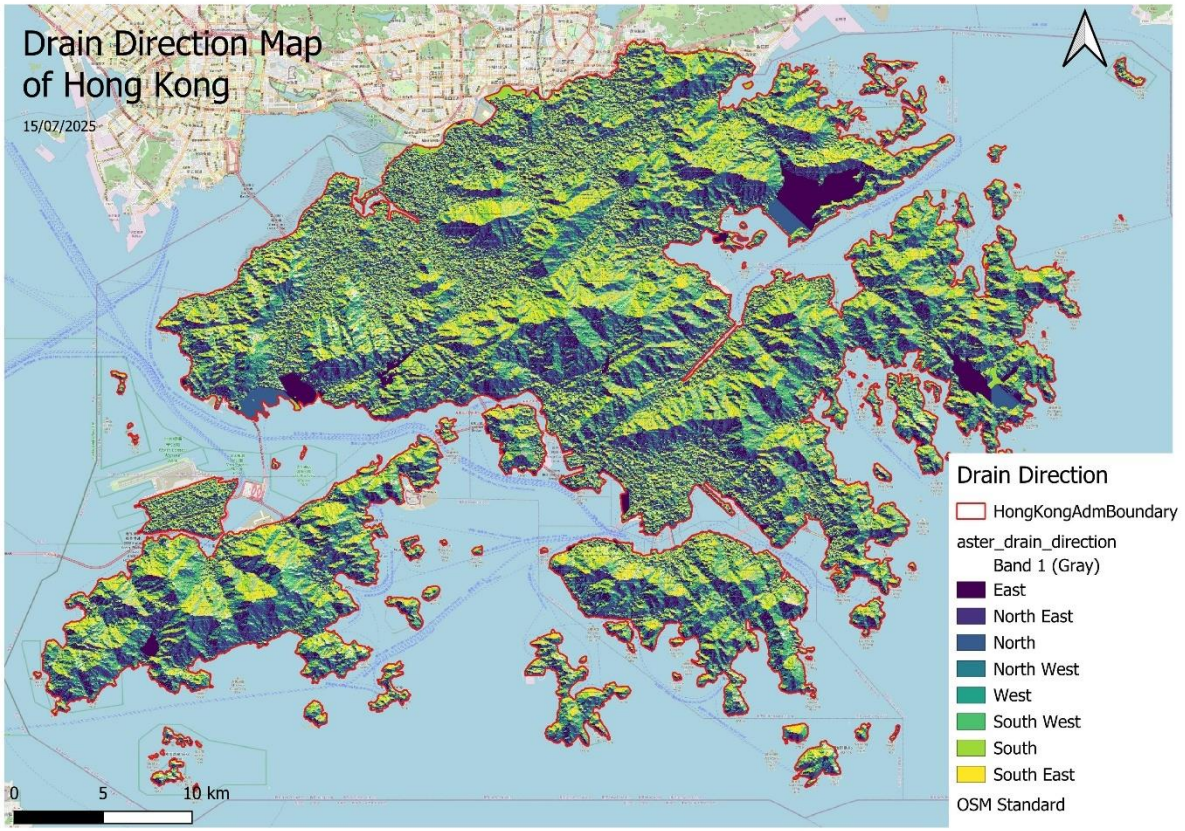
ExposedRailway_PFZ
PersistentFloodingZones
HongKongAdmBoundary
hotosm_hkg_railways_lines_shp
OSM Standard

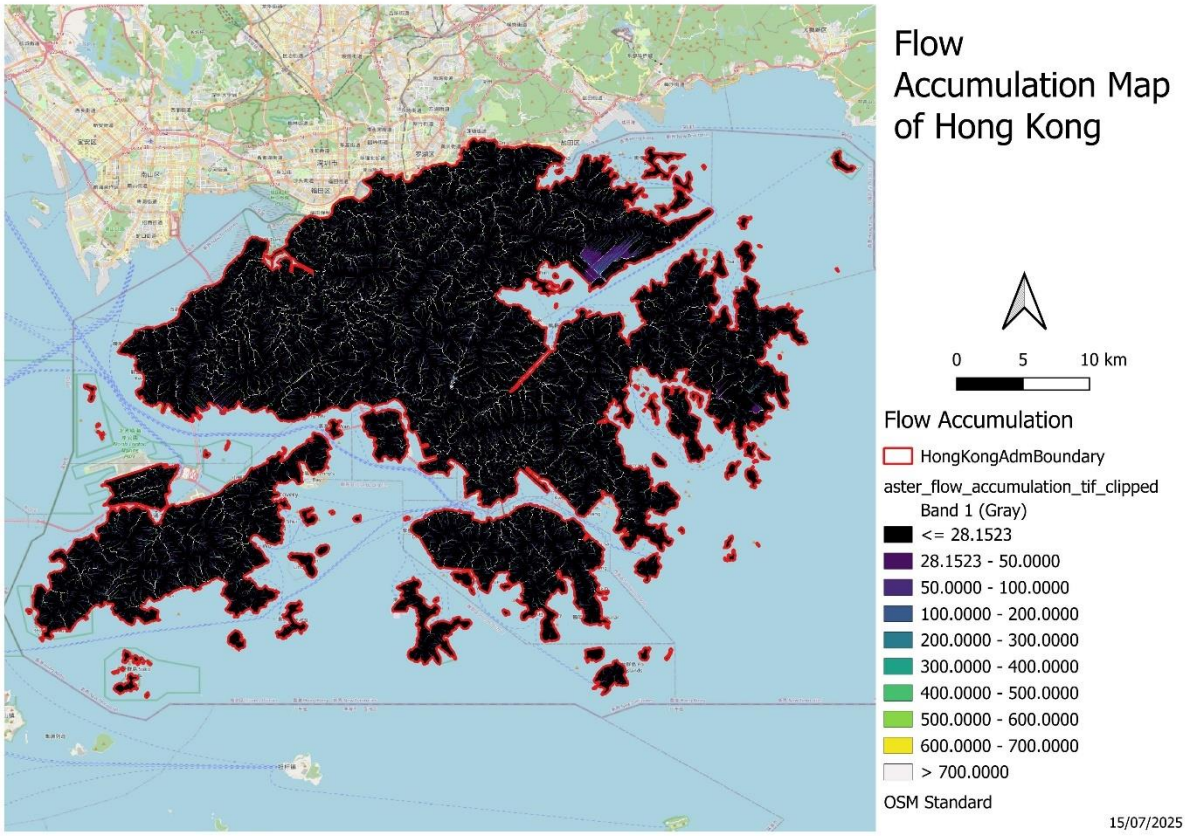
15/07/2025

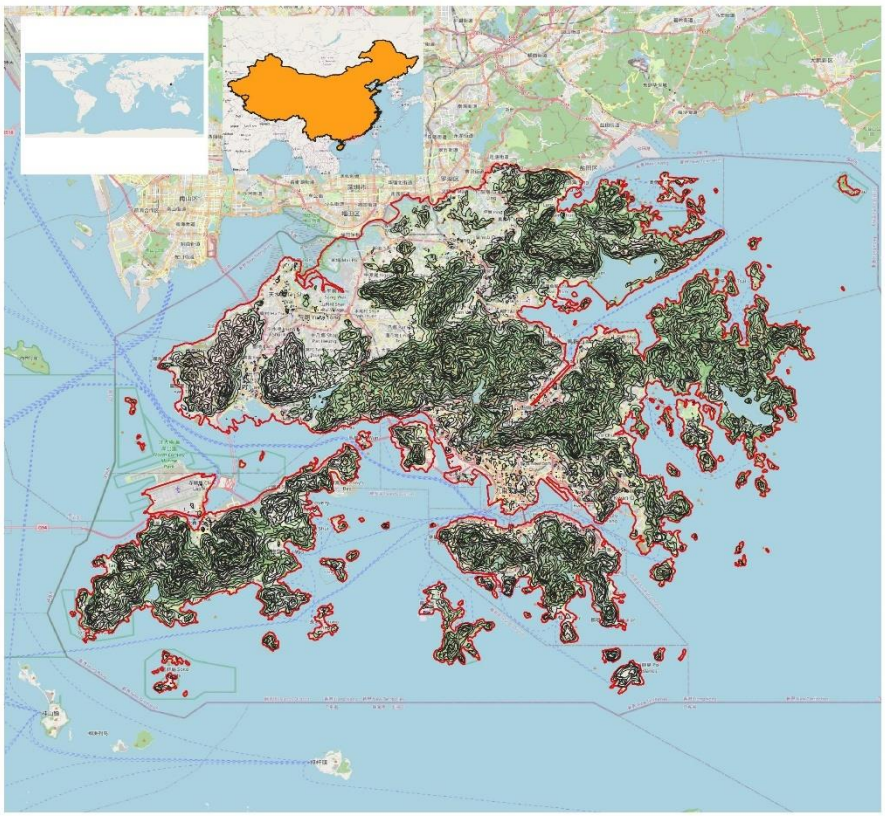
5. Terrain Map (Hillshade + Contours)






Drain Direction Map
of Hong Kong
15/07/2025
0
5
10 km
Drain Direction
HongKongAdmBoundary
aster_drain_direction
Band 1 (Gray)
East
North East
North
North West
West
South West
South
South East
OSM Standard


Flow
Accumulation Map
of Hong Kong
0
5
10 km
Flow Accumulation
HongKongAdmBoundary
aster_flow_accumulation_tif_clipped
Band 1 (Gray)
<= 28.1523
28.1523 - 50.0000
50.0000 - 100.0000
100.0000 - 200.0000
200.0000 - 300.0000
300.0000 - 400.0000
400.0000 - 500.0000
500.0000 - 600.0000
600.0000 - 700.0000
> 700.0000
OSM Standard
15/07/2025



Contour Map of Hong Kong

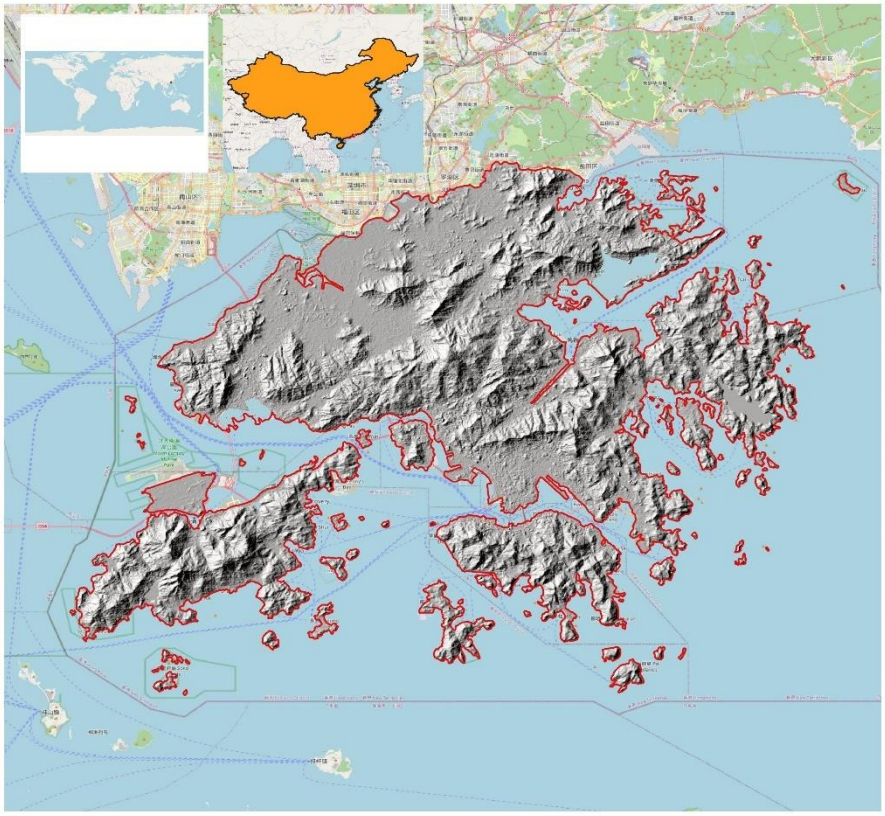
0 5 10 km

Contours_30m

Contours

HongKongAdmBoundary

OSM Standard



Hillshade Map of Hong Kong

0 5 10 km

Hillshade

HongKongAdmBoundary

hillshade

Band 1 (Gray)

244.865

93.8076

OSM Standard

Key Findings

1. **Flood-Prone Areas Identified**
    - Using slope (<5°) and flow accumulation (>1000 cells), flood risk zones in Hong Kong were delineated.
    - These zones were polygonized and visualized for further analysis.
2. **Infrastructure Vulnerability Quantified**
    - Railway line exposure to flood zones was assessed.
    - **Impacted Length**: ~12.13 km
      Total Length: ~338.72 km
      % Impacted: **3.58%** of railway network lies in potential flood zones.
    - Education and Health facilities were also assessed.
3. **High-Accuracy Supervised Classification**
    - Supervised LULC classification using Sentinel-2 stack and manually labeled training data achieved.
4. **Zonal Statistics Generated**
    - Land cover distribution was calculated within flood-prone polygons (though class-wise stats were mostly null due to coarse vector geometry or label mismatch—this can be refined).
5. **Fully Automated Workflow**
    - Pipeline includes - DEM clipping → hydrological modeling → flood zone extraction → infrastructure intersection → Sentinel-2 mosaicking → unsupervised and supervised classification → zonal statistics.

Limitations

1. **DEM Resolution Dependency**
    - Hydrological modeling and flood extent accuracy depend on DEM quality and resolution.
    - Coarse DEM may smooth out critical terrain features affecting local drainage.
2. **Threshold-Based Flood Modeling**
    - Simple slope and accumulation thresholds were used to simulate flood zones.
    - Does **not include rainfall, infiltration, soil type, or river capacity**—making it a basic proxy for risk zones.
3. **Training Sample Quality**
    - Supervised classification depends on spatial accuracy and representativeness of training samples.
    - Less number of ROIs can reduce performance.
4. **Memory Constraints**
    - Large tile sets caused memory errors.
    - Scalability is limited by local system resources.
5. **Limited to Sentinel-2**
    - Optical-only classification; currently **no SAR or thermal** bands were integrated. Hence, flood detection or permanent waterbody classification may be limited in cloudy conditions.

Assumptions

1. **Uniform Rainfall and Flow Direction Assumed**
    - The use of D8 accumulation assumes water flows in only 1 of 8 directions—oversimplifying complex flow behaviors.
2. **Static Infrastructure and Land Use**
    - LULC and railway data are treated as static snapshots, not accounting for seasonal or temporal variability.
3. **Flood Thresholds are Generic**
    - The values (e.g., 1000 cells for flow accumulation) are not calibrated with historical flood data.
    - Buffer distances for flood risk zones are generic.